

When the Scaffold Stays On: AI, Practice Style, and Screening in Elite Skill Formation

Song Yao*

June 3, 2026

Abstract

Generative AI raises short-term productivity by completing tasks that learners would otherwise practice on their own. Whether this substitution erodes frontier skill, the skill required for top-tail non-AI-aided performance, is an open question with growing stakes wherever AI is unavailable or insufficient. The sharper question, however, is whether selection mechanisms can screen apart two coexisting types: substitute-users, who use AI in place of deliberate practice, and complement-users, who use it to accelerate skill development. We exploit a feature of elite programming. The International Collegiate Programming Contest (ICPC) and the International Olympiad in Informatics (IOI) prohibit AI under in-person proctoring and admit entrants through qualification rounds, whereas online Codeforces (CF) contests are unproctored and open to all. From CF submission histories we build an AI-prompt signature, marked by more first-attempt acceptances and fewer attempts and debugging retries, consistent with AI-assisted practice. Three patterns triangulate an institutional-screening mechanism. First, CF practice has shifted toward this signature across entry cohorts spanning two AI rollouts. Second, in open CF contests a stronger signature predicts smaller CF rating gains for users with no ICPC or IOI affiliation, but not for users who qualified for those AI-prohibited contests. Third, inside the AI-prohibited ICPC environment, a shift toward this AI-style practice predicts higher non-AI-aided scores for AI-era entrants. The same practice input thus carries opposite signs depending on whether the contest environment screens for it. The contrast points to two levers: how AI is integrated into training, since within the screened pool AI-style practice coincides with stronger non-AI-aided performance; and the design of AI-prohibited evaluation gates as a type-separating institution. Both extend beyond competitive programming to credentialing systems, from medical and legal boards to professional certification, that must certify skill in a workforce whose practice is increasingly shaped by AI.

Keywords: generative AI; human capital; skill formation; frontier skill; substitutes and complements; screening; selection mechanisms; credentialing; competitive programming.

*Olin Business School, Washington University in St. Louis. Email: songyao@wustl.edu.

1 Introduction

Generative AI can raise short-term output by handling work that learners would otherwise do on their own. Whether that substitution reduces learners’ investment in deep, hard-won human capital, the kind required for top-tail *non-AI-aided* performance, is a question with growing stakes, especially for frontier work that requires creativity and ability beyond what AI can provide, or where AI is unavailable. We reframe the question on two margins. First, does AI exposure during training predict differential later performance in AI-aided versus AI-prohibited settings? Second, can selection mechanisms distinguish learners who use AI as a substitute for deliberate practice from those who use it to enhance their skill development?

Together these two questions are sharper than “does AI affect skill?” For example, calculators make many forms of skill loss look benign: if a tool is always available and reliable, the underlying capability matters less. That argument applies to *routine* skill but breaks at the frontier. AI fails on genuinely novel problems by design (it recombines patterns from its training data); catching AI errors requires non-AI-aided expertise; trusted institutions such as peer review, medical diagnosis, judicial reasoning, and scientific reproducibility need people who can function without AI mediation; and the innovation pipeline that feeds future training data depends on humans pushing the frontier. Frontier skill is what AI cannot substitute for. By displacing the deliberate practice required for human capital accumulation, AI puts that very frontier skill under pressure.

The mechanism mirrors, at the individual level, the macro substitution dynamic in [Acemoglu et al. \(2026\)](#): AI substitutes for human effort and weakens the human-capital investment that produces frontier capability. AI affects skill formation by changing how learners practice. That change in practice effort is what we aim to measure, rather than noise we set aside. We frame the analysis as an extension of standard dynamic skill-formation models ([Cunha et al. 2010](#), [Becker 1964](#), [Mincer 1974](#)) in which AI exposure differentially affects practice effort across two coexisting subpopulations: *substitute-users*, who let the AI do the work and whose underlying skill therefore fails to develop, and *complement-users*, who use AI as a learning accelerator while continuing to invest in human capital through deliberate practice. The substitute-versus-complement distinction reflects a broader pattern: the same technology produces heterogeneous usage patterns across people with identical access, and the design of credentialing institutions in the presence of such heterogeneity is a mechanism-design problem with crucial socioeconomic stakes. Our empirical strategy in [Section 5](#) identifies this heterogeneity by exploiting the selection structure of AI-prohibited contests: AI-prohibited qualification gates filter out substitute-users, leaving complement-users in the elite pool.

We test these ideas in elite competitive programming, a setting that pairs an open, AI-permissive training platform with AI-prohibited contests. Codeforces (CF) is an online platform *open to all*:

its contests have no qualification filter, so the pool blends substitute-users and complement-users, and AI use during practice is unregulated. Individual CF practice submission histories provide a long panel of submission-level detail on how users practice; CF contest outcomes, on the other hand, provide continuous individual performance measurements. The International Collegiate Programming Contest (ICPC) and the International Olympiad in Informatics (IOI) sit at the other extreme: both prohibit AI and run qualification rounds that screen entrants before the finals (ICPC through peer screening at team formation and team qualifiers, IOI through national olympiads), so the qualifying pool is filtered toward complement-users. Our panel follows 10,419 CF users across two AI rollouts, the GitHub Copilot launch in 2021 and the ChatGPT release in 2022; a subset of these users are linked to ICPC and IOI rosters and contest scores.

Three patterns emerge from the analyses on these users, and they differ in how much identifying weight they carry. We present them in that order: a descriptive shift in practice norms across entry cohorts, a suggestive but bounded cross-subpopulation asymmetry, and a robust within-contest association that anchors the paper’s central claim.

A shift in practice norms across entry cohorts (Prediction 1). Across the platform, the CF *practice-mode* submissions of later entry cohorts lean toward a behavioral signature: more problems solved on the first attempt, fewer attempts per solved problem, and fewer rapid debugging retries. We call this an *AI-prompt signature* because it matches what a prompt-and-fix workflow with an AI assistant would produce.¹ To illustrate with one of the three components, the fraction of problems solved on the first attempt, the AI-era cohort’s relative post-ChatGPT change is 0.0586 (SE 0.0063, $p < 0.001$) above the pre-AI cohort, and it is present in both the ICPC-IOI-rostered (0.0356) and non-rostered (0.0617) subpopulations rather than driven by either pool alone; the other two components shift in the same AI-prompt direction. We treat this cohort contrast as descriptive context: it concerns practice inputs, it does not by itself separate substitution from complementarity, and because the shift is not cleanly timed to the AI releases (Section 5.1) we do not read it as AI-caused.

A cross-subpopulation asymmetry in the open pool (Prediction 2). On CF rating, which measures CF contest performance, a stronger AI-prompt signature during practice predicts smaller rating gains among non-rostered users (-12.24 points per signature unit, $p < 0.001$) but is uncorrelated with rating change among ICPC-IOI-rostered users (-2.41 , $p = 0.58$), who passed the AI-prohibited gates of ICPC or IOI qualifying rounds. The cross-subpopulation difference is -9.82 ($p = 0.048$, 95% confidence interval $[-19.58, -0.07]$; $N = 6,690$, comprising 5,283 non-rostered and 1,407 roster-linked users). The sign is what the screening mechanism predicts — the signature tracks rating loss where substitute-users remain in the pool but not where the gate has removed them. But the result is suggestive rather than decisive: it sits at the conventional significance threshold

¹The signature is a practice-style proxy consistent with AI-assisted practice rather than a verified measure of AI use.

and, under differential attrition between the two subpopulations, the selection bounds reach near zero in the worst case (Section 5.2).

A positive AI-intensity slope inside the screened pool (Prediction 3). Inside the AI-prohibited ICPC environment, a within-user shift toward the AI-prompt practice style predicts *higher* non-AI-aided contest score for users who entered competitive programming in the AI era. Regressing ICPC score on this practice-style change (the first principal component of the three signature components) interacted with cohort, the AI-era cohort interaction is 0.190 ($p = 0.028$, 95% confidence interval [0.020, 0.360]; $N = 579$ contest-years across 444 users), while the pre-AI cohort slope is a null -0.048 . This within-contest association is the anchor result for the paper’s central claim: it is identified inside the AI-prohibited environment, is specific to the AI-era cohort, and survives career-stage and baseline-skill controls (Section 5.3); its magnitude is construct-dependent, so we read it as establishing the sign of the effect rather than its precise size. Once the gate has screened substitute-users out, the AI-style practice that remains tracks complement use that raises performance. Predictions 2 and 3 together are the sharpest test, since the same practice input carries opposite signs depending on whether the contest environment screens substitute-users out.

We draw three lessons, leading with the one the evidence supports most directly. First, AI-prohibited qualification gates act as a population-level filter that separates substitute-style from complement-style practice, sharpening the classical signaling and screening framework of credentials (Spence 1973, Stiglitz 1975) for a setting in which the underlying skill is itself partially substitutable by AI. The opposite-signs pattern across screened and unscreened environments is the evidence for this filtering. Second, the relevant object is how learners use the tool rather than whether they can access it: the same practice input predicts performance loss in the open pool and performance gain inside the screened pool. Third, the findings point to two policy levers. The evidence most directly supports gate design: AI-prohibited evaluation gates at credential boundaries, already standard at elite programming contests. The framework motivates a second lever, redesigning training to preserve non-AI-aided practice alongside AI. Whether either lever generalizes to other elite credentials depends on how far AI substitutes for skill-building practice in those domains.

The remainder of the paper is organized as follows. Section 2 positions our contribution against the existing literature on AI and skill formation. Section 3 describes the AI shocks, the competitive-programming setting, and the testable predictions. Section 4 describes the data. Section 5 presents the empirical strategy and results. Section 6 discusses mechanisms, threats to identification, and implications before concluding.

2 Related Literature

This paper sits at the intersection of five literatures. First, macroeconomic theory provides a mechanism of human capital depreciation when AI substitutes for human effort. We extend the framework by distinguishing substitute-users from complement-users of AI and test the resulting predictions at the individual level. Second, randomized controlled trials (RCTs) on AI-assisted learning establish the short-term substitution mechanism; the question is whether the mechanism persists in long-term skill formation. Third, workplace and educational observational studies document how AI changes practice and production patterns at scale; we extend those patterns into an elite skill market with a multi-year time window. Fourth, pre-generative-AI evidence from chess training engines shows distributional compression — lower-tail player ratings rise while the world-champion rating stays flat — under complement-style tool use; the open question is whether the elite frontier erodes when the AI can do the work itself, which the chess setting does not test. Finally, we frame AI-prohibited credentialing as a mechanism-design problem, drawing on signaling-and-screening theory in economics. We discuss each stream in turn.

2.1 Macroeconomic theory of AI and human capital

[Acemoglu et al. \(2026\)](#) provide a dynamic theoretical model in which agentic AI substitutes for human effort, weakening the flow of new knowledge people could contribute to society’s general knowledge stock. Consequently, general knowledge depreciates. Furthermore, under certain conditions, if AI accuracy rises above a critical threshold, complete knowledge collapse becomes the unique stable outcome because people stop investing any effort and delegate work to AI. The model is macro-level, theoretical, and assumes symmetric agents. We test this depreciation mechanism empirically at the individual frontier level, and relax their symmetry assumption: AI exposure differentially affects practice effort across substitute-user and complement-user subpopulations, with the heterogeneity identified through selection into AI-prohibited contests (Section 2.5). Together these provide observational evidence of AI’s effects on long-run skill formation in an elite skill market.

2.2 Substitution mechanism evidence from RCTs on AI-assisted learning

Recent RCTs have tested AI’s substitution-for-effort mechanism in short-horizon settings. [Stadler et al. \(2024\)](#) find that students using ChatGPT for a 20-minute research task expended about 35% less germane cognitive load (the deep-processing facet that builds understanding) than students using Google search, and produced worse output as a result. [Melumad and Yun \(2025\)](#) show across seven experiments that learning from LLM summaries yields shallower knowledge than learning from web

search, because the model performs the synthesis the user would otherwise do themselves. Closest to our design, [Shen and Tamkin \(2026\)](#) randomized 52 software developers learning a new Python library: those with high-engagement patterns (asking conceptual questions, requesting explanations alongside generated code) matched or exceeded non-AI users on an unaided follow-up quiz, while those with low-engagement patterns (debug delegation) scored lower. Our *AI-prompt signature* on CF submissions is the multi-year, field-observational analog of their low-engagement pattern.

[Bastani et al. \(2025\)](#) provide the cleanest existing test of substitution-erosion: their field RCT randomizes AI access across 1000 students during mathematics practice and then evaluates their performance on AI-free in-class exams immediately after each practice session. The students were randomized across three arms (Control with no AI access, “GPT Base” directly giving answers to questions, “GPT Tutor” giving hints for solving questions). GPT Base raised practice grades by 48% but reduced exam grades by 17%; GPT Tutor raised practice grades by 127% without harming exam grades. In short, because GPT Base students use AI as substitute for problem-solving during practice, their higher practice grades did not carry through to the AI-free exam.

These RCTs establish substitution at the session and task scale; whether it persists is our question. Four features separate our setting. [Bastani et al. \(2025\)](#) run four 90-minute sessions, whereas we follow contestants across multiple years (2018–2025); they estimate mean effects in high-school mathematics, whereas we estimate top-tail effects in elite competitive programming; their AI-prohibited test is a high school course exam, whereas ours are career-defining contests (CF, ICPC, IOI) feeding university admissions and employer signals; and they study students working below the frontier, whereas we study the formative-to-elite pathway in a domain where the frontier is what AI is least able to substitute for. We test the multi-year, top-tail, real-stakes joint cell these dimensions define.

2.3 Workplace and observational AI studies

A growing observational literature documents how AI tools change practice patterns in workplaces and learning environments. [Brynjolfsson et al. \(2025\)](#) document heterogeneous productivity gains in a call-center setting consistent with distributional compression: AI raises mean output, with lower-skilled workers gaining more than higher-skilled ones. [Hoffmann et al. \(2024\)](#) show that GitHub Copilot shifts developers’ time toward core coding and away from project management, with larger effects for lower-ability developers. [Shihab et al. \(2025\)](#) document a programming workflow shift from read → understand → implement to prompt → view response → implement under Copilot, with manual code entry dropping from 95.6% to 52.5%. [Hausman et al. \(2025\)](#) show that ChatGPT raises university grades, especially for lower-performing students, compressing the grade distribution and eroding grades’ signal value for employers.

These studies show that AI changes how people work and learn. Their scale is session-level or within-firm; the time window is months. We extend the inquiry to multi-year individual horizons and elite-frontier outcomes. The 2018-2025 CF panel follows individual contestants across two major AI rollouts, and the subpopulation who also appear in ICPC and IOI rosters links their AI-permitted practice to AI-prohibited contest performance.

2.4 Pre-generative-AI evidence: chess engines and why they differ from AI in programming

Gaessler and Piezunka (2023) document that pre-generative-AI computer-assisted training in chess raised players’ annual Elo ratings (a relative skill rating based on chess game outcomes) by approximately 11 points per year on average. The effect concentrated in lower-skilled players, with world-champion-level Elo being flat during the same period. The chess engines’ impact is therefore a population-level distributional compression: the lower tail rises, the frontier holds steady.

This positive Elo result is the most natural objection to our framing. Computer engines have been part of chess training for decades, raising the lower tail while leaving the frontier intact; if the tool did not erode the chess frontier, why would AI-assisted training in programming threaten deep-skill formation at all? The answer turns on three concrete differences between the chess-engine training regime and current AI-programming practice. Each difference moves chess closer to a complement-style integration and AI-programming closer to a substitute-style integration.

1. **Complement versus substitute.** A chess player consults an engine as an oracle during analysis: it answers “what is the best move here?”, but the player still has to internalize *why*, and the engine never plays the game. The engine’s output is feedback, not output. AI for programming integrates into the work itself — prompt, generate, accept or modify — and can deliver functioning code without the user grasping the algorithm. Stadler et al. (2024) measure the difference at the cognitive level: AI use suppresses germane cognitive load (the deep processing that builds skill) by 35%. Chess engines do not produce comparable suppression because they do not substitute for that processing.
2. **Training-test separation.** Competitive chess maintains a hard institutional boundary: engines are forbidden during competition, so players know precisely when they are using the tool and when they are competing unaided. Everyday programming has no equivalent boundary; AI is integrated continuously, with no natural moment that says “stop using AI now.” Bastani et al. (2025) operationalize the distinction: GPT Tutor (scaffolded, no full solutions) did not harm unaided performance, while GPT Base (continuous full-solution access) cut unaided exam scores by 17% — same domain, same students, scaffolded versus unscaffolded use. AI-prohibited

contests (Section 2.5) reinstate the chess-style separation that everyday programming lacks, and that separation is the feature our research design exploits.

3. **What the tool can substitute for.** A chess engine is a powerful study aid, but unaided games assess runtime cognition the engine cannot perform in the moment: pattern recognition under time pressure, calculation, strategic planning, composure. There is a gap between what the engine does (analysis between games) and what competition measures (live play). AI in programming substitutes for design-time cognition — algorithm design, debugging, problem decomposition — which is precisely what programming contests measure. There is no such gap: the AI does the very thing the contestant is scored on.

Read together, these differences make the positive Elo result of [Gaessler and Piezunka \(2023\)](#) consistent with — not against — our framing: chess engines raised ratings through a complement-style mode of use, whereas current programming practice sits closer to the substitute side. Whether this substitute-style risk shows up in an elite skill market over multi-year horizons is the question this paper tests.

2.5 Selection into AI-prohibited credentialing

The previous four streams establish what AI does to skill formation through substitution mechanisms and what it does to practice patterns at workplace scale. The fifth question motivates our research design: can evaluation institutions screen productive from unproductive AI use when the technology partially substitutes for the skill being evaluated?

The distinction we draw is between substitute-users and complement-users of AI: contestants with the same AI access during practice differ in whether they use it as a substitute for deliberate practice or as a complement, and that difference accumulates into divergent non-AI-aided performance. This is the multi-year, field-scale analog of the engagement-pattern heterogeneity in the RCT evidence (Section 2.2): the same tool yields different skill outcomes depending on how the user engages it.

The evaluation gate filters through two channels at once: formal rules (no AI, no internet, in-person, supervised) and social selection (peer screening at team formation, national-olympiad reputation), which together determine who reaches the elite qualifying pool. The design question is not whether to permit AI but how to make the gate filter on the substitute-vs-complement margin. Classical signaling ([Spence 1973](#)) and screening ([Stiglitz 1975](#)) theory treat credentials as costly actions or tests that separate types; our setting differs in that the effort is genuinely skill-forming rather than purely type-sorting, and the institution actively shapes which separating equilibrium obtains by designing the gate. What is new with generative AI is that the deep skill is now partially substitutable by AI, so the gate's filtering structure, not just its existence, determines whether the credential still separates types.

We make three contributions to this stream: observational evidence at multi-year individual scale that the substitute-vs-complement distinction is empirically real and accumulates into divergent non-AI-aided performance; identification of a selection mechanism, in which ICPC’s peer screening at team formation and IOI’s national-olympiad filter screen substitute-users out before the contest, which we treat as a research-design feature rather than a sample-selection nuisance; and a sharpened policy claim that gate design should target the substitute-vs-complement margin rather than rely on generic gatekeeping. Together these position the contribution as a design principle for evaluation systems in the presence of partially substitutable AI.

3 Background and Empirical Setting

3.1 The AI shocks

GitHub Copilot entered technical preview on June 29, 2021 and reached general availability one year later. It targeted programming directly and was among the first AI tools widely adopted by programmers. ChatGPT followed on November 30, 2022, putting generative AI into broad public use across populations and tasks. Subsequent releases (e.g., GPT-4, Claude 2 and 3, DeepSeek-Coder) further extended AI’s coding capability but did not introduce a comparably sharp adoption shock.

The two tools also embody different functional affordances ([Markus and Silver 2008](#)), the possibilities for action a technology offers its users. The in-editor autocomplete of Copilot affords inline, line-level code suggestion that a programmer accepts or edits while typing; the conversational interface of ChatGPT affords whole-solution generation, natural-language explanation, and iterative debugging through prompting. These differing affordances shape how AI is enacted in routine practice ([Orlikowski 2000](#)), and they help interpret why the quick-retry margin shifts in opposite directions around the two releases (Section [5.1](#); Appendix [A.2](#)).

The two releases play different roles in our research design. Copilot was a programmer-specific tool and arrived more than a year before ChatGPT, so users who first appeared on CF after the Copilot release had already begun training with AI tools available to them. ChatGPT was a broader population-level event with a sharper date, observable across the entire panel rather than only among the early adopters of Copilot. We use the Copilot date to partition users by training vintage, and the ChatGPT date to define the within-user pre/post comparison; we elaborate on the motivation for this split in Section [5](#).

3.2 Competitive programming as the empirical setting

CF is the dominant online competitive programming platform. It has run contests several times a week since 2010. It issues individual ratings based on each user’s performance in the contests compared to others. The CF rating is Elo-style, like chess, measuring each user’s *relative* skill. Besides weekly contests, anyone can create an account and submit practice solutions. Practice submissions do not enter the rating system and are a good training sandbox. There is no qualification gate, no proctoring during practice, and nothing preventing a user from running an AI tool while solving problems. The platform is, in this sense, an open pool during the practice phase: the participating population mixes substitute-users with complement-users, and the platform does nothing to separate them. During rated contests AI use is again hard to police, but AI may be a weaker substitute there: under time limits and a point penalty for incorrect submissions, contestants who lean on AI without the underlying skill tend to lose rating rather than gain it (see Section 5).

The International Collegiate Programming Contest (ICPC) and the International Olympiad in Informatics (IOI) sit on the other end of the institutional spectrum. Both run multi-stage qualification rounds before the finals, and both prohibit AI tools at every stage. ICPC builds teams of three from a single university and holds its AI-prohibited qualification rounds and World Finals in person. IOI selects individual contestants through national olympiads and runs the international round in person (online only during 2020 and 2021 due to COVID-19). Users whose CF performance leans heavily on AI are unlikely to pass these rounds, so the populations that reach the ICPC finals and the IOI delegations have been screened of the most AI-dependent practitioners. These are high-stakes contests whose medals are recruiting signals and are cited on university applications, so the non-AI-aided scores are a meaningful frontier measure rather than a lab proxy.

The contrast between the more AI-lenient CF pool and the screened ICPC and IOI pools is what lets us identify the substitution mechanism. The CF pool keeps both user types visible and shows how AI exposure correlates with rating change in an unscreened population. The qualified ICPC and IOI pools have been pre-filtered by an AI-prohibited gate and show how AI-intensive practice translates into non-AI-aided performance among the survivors of that filter. Reading the two together gives cross-environment evidence on the same training inputs.

On the CF platform, one further feature matters for our empirical strategy. Each submission carries a tag indicating whether it is in practice or contest mode. The behavioral signature of AI-style practice is cleanest in practice mode, where substitution operates freely and the data are dense, so we use the practice-mode signature as our main measure of what we term *AI-intensity*: a practice-style proxy for how far a user’s practice has shifted toward an AI-style prompt-and-fix workflow, although not a verified measure of AI use. The same signature computed on contest-mode submissions, noisier because contest behavior also reflects contest skill and time pressure, provides a robustness check against a roster-versus-skill confound (Appendix A.3): the cross-subpopulation

pattern it shows is consistent with the practice-mode measure.

3.3 Testable predictions

The setting described above pairs an open, AI-permissive practice environment (CF) with AI-prohibited contests (ICPC and IOI) whose qualification rounds screen entrants. Two features of this structure generate the predictions that the rest of the paper tests. The first is the substitute-versus-complement distinction of Section 2.5: users with the same AI access during practice differ in whether they let the tool do the work, so that their non-AI-aided skill does not develop (substitute-users), or use it as a learning accelerator while continuing to invest in deliberate practice (complement-users). The second is selection: because an AI-prohibited gate cannot be passed on substituted skill, it filters substitute-users out of the qualified pool, leaving a screened population tilted toward complement-users.

These two features yield three predictions, which Section 5 tests in turn.

Prediction 1 (universal practice shift). After the AI releases, CF practice behavior shifts toward an AI-prompt signature — more first-attempt acceptances, fewer attempts per solved problem, and fewer rapid debugging retries — across the platform as a whole, and the shift grows with how much of a user’s training postdates AI. This prediction concerns practice inputs and does not by itself separate substitution from complementarity.

Prediction 2 (asymmetry in the un-screened pool). In the open CF pool, where both user types coexist, a stronger AI-prompt signature predicts smaller rating gains among un-screened (non-rostered) users, because substitute-users remain in this pool and their non-AI-aided skill erodes. Among roster-linked users, who have passed AI-prohibited gates, the same signature does not predict rating loss, because the gate has already removed the substitute-users. The signature-to-rating slope is therefore more negative in the un-screened subpopulation than in the screened one.

Prediction 3 (positive slope inside the screened pool). Inside the AI-prohibited ICPC environment, among users who entered competitive programming in the AI era, a larger shift toward AI-intensive practice predicts *higher* non-AI-aided contest score. Once the gate has screened substitute-users out, the AI-style practice that remains reflects complement use, which builds frontier skill rather than eroding it.

Predictions 2 and 3 together form the sharpest test. The same training input, a shift toward AI-style practice, is predicted to carry *opposite* signs depending on whether the contest environment screens substitute-users out: negative in the un-screened CF pool (Prediction 2) and positive inside the screened ICPC pool (Prediction 3). A single mechanism operating uniformly, whether pure erosion or pure enhancement, cannot produce both signs; only the combination of heterogeneous use and selective screening does.

4 Data

4.1 Data Sources

The analysis draws on three data sources, listed in Table 1. The two AI-prohibited contests, ICPC and IOI, have public rosters that identify the credentialed competitors who anchor the CF panel. Specifically, the website Competitive Programming Hall of Fame (cphof.org) supplies ICPC rosters and team scores from 2018 through 2025; IOI’s website (stats.ioinformatics.org) hosts IOI individual scores over the same time window. The CF individual panel is then assembled from the CF API in stages: we link CF accounts to the ICPC and IOI rosters and scrape those rostered competitors, then add a non-rostered expansion, resulting in the full CF rating and submission histories of 10,419 CF users. Next, we detail the scraping process and the linkage- and cohort-window exclusions that define this analysis panel.²

Table 1: Data sources and roles.

Source	Records	Outcome type	Role
<a href="http://cphof.org/standings/icpc/<year>">cphof.org/standings/icpc/<year>	1,055 team-year records; 3,163 contestant-year records (2018–2025)	Team-level problems solved, ranks, and medals	ICPC AI-prohibited test events; CF user-level linkage via team rosters
<a href="http://stats.ioinformatics.org/results/<year>">stats.ioinformatics.org/results/<year>	2,779 contestant-year records (2018–2025)	Individual-level points, ranks, and medals	IOI AI-prohibited test events; CF user-level linkage via country delegation rosters
CF API	10,419 users; 793,920 user-month records; 2018–2026	CF rating + submission verdicts + submission timestamps + profile	Within-individual panel, assembled from the scrapes detailed in Section 4.2; user linkage to ICPC/IOI rosters

4.2 Panel Construction

The analysis uses two unbalanced panels: the panel from the AI-prohibited ICPC and IOI contests, and the CF within-individual panel.

²We considered the larger public CF submissions dataset on Hugging Face ([denkCF/UsersCodeforcesSubmissionsEnd2024](https://huggingface.co/datasets/denkCF/UsersCodeforcesSubmissionsEnd2024); 17.6 million submissions, 14,957 anonymized users), but its users are anonymized and cannot be matched to ICPC or IOI rosters; we therefore scrape real, identifiable users directly from the CF API.

The AI-prohibited panel comprises 1,673 distinct ICPC contestants and 1,030 IOI contestants. Together they result in 2,494 distinct ICPC-IOI rostered contestants (209 competing in both) whose first rated CF contest falls between 2015 and 2024. The Competitive Programming Hall of Fame (cphof.org) publishes the CF usernames for participants of ICPC finals if they have a CF account. For IOI contestants (stats.ioinformatics.org), we use name-and-country matching to find their CF accounts. Consequently, we are able to match each of these 2,494 contestants to their respective CF account, forming the ICPC-IOI roster-linked subpopulation. These ICPC and IOI contestants span 354 universities and 90 countries over 2018–2025. Panel D of Table 3 documents the full ICPC-IOI-CF record linkage, which covers 3,117 roster-linked CF accounts across all cohorts (236 linked to both ICPC and IOI); the 2,494 contestants here are the cohort-assignable subset whose first rated CF contest falls in the 2015–2024 window, and the remaining 623 fall outside that window and are not assigned a cohort.

The CF within-individual panel contains 10,419 users, of whom 6,399 (61.4%) span both the pre-Copilot window (on or before 2021-06-28) and the post-ChatGPT window (on or after 2022-11-30) with at least 30 practice submissions on each side. These 10,419 users come from two groups. The first is 2,480 ICPC-IOI roster-linked users (from the 2,494 linked contestants above, less 14 with no practice-mode submissions in the analysis window; see footnote). The second and larger group is a non-rostered expansion of 7,939 users who are not anchored to either ICPC or IOI roster. These non-rostered users are scraped directly from the CF API among active accounts with a maximum rating of at least 1,400; these non-rostered users make up roughly three-quarters of the panel. The rating floor applies only to the non-rostered expansion; the non-rostered findings therefore speak to the moderately-skilled-and-above CF population (maximum rating $\geq 1,400$) rather than to the full platform.³

Every user in the CF panel is assigned to one of three cohorts according to the date of their first rated CF contest, following the cohort structure in Table 2: *pre_AI* (2015 through 2018), *transition* (2019 through 2021-06-28, the day before the Copilot launch), and *AI_era* (2021-06-29 through 2024). The rationale for the three-cohort structure is to compare programmers by their skill maturity relative to AI availability. Accordingly, the cohort split is by their training vintages relative to the

³The analysis panel excludes three groups of scraped users. First, 1,374 users are matched to an ICPC or IOI roster only by a lower-confidence name-and-country match; because the match is too uncertain to confirm their participation and attach a contest score, we keep only their CF practice histories and exclude them from the analysis panel, and Appendix A shows the within-subpopulation slopes are insensitive to their handling (excluded, assigned to the non-rostered subpopulation, or assigned to the roster-linked subpopulation); only the pooled interaction softens, to $p = 0.110$, when these AI-era-heavy matches are forced into the roster-linked arm, which is itself a reason to prefer their exclusion. Second, 1,408 users cannot be assigned a cohort because their first rated CF contest falls outside the 2015–2024 cohort window: 623 rostered contestants (beyond the 2,494 rostered contestants above) and 785 non-rostered users. Third, 38 users have no practice-mode submissions in the analysis window of 2018-Q1 through 2025-Q4 (excluding the 2022-Q4 boundary quarter): 14 roster-linked (contest-only or pre-2018 practice activity) and 24 non-rostered. Beyond these three exclusions, the panel is not filtered on any outcome variable.

Table 2: Cohort definitions.

Cohort	First-rated-contest window	AI availability during training
pre_AI	2015-01-01 to 2018-12-31	Trained entirely pre-Copilot. Reference group.
transition	2019-01-01 to 2021-06-28	Partly pre-AI; AI tools arrived mid-career.
AI_era	2021-06-29 (Copilot launch) to 2024-12-31	Trained with AI tool availability.

Copilot launch on 2021-06-29: a pre_AI user had largely trained before AI, a transition user was halfway through training when AI arrived, and an AI_era user trained with AI available throughout. We then observe the panel’s CF practice and contest outcomes from 2018-01-01 to 2026-04-30.

4.3 Outcome measurement: scoring system properties

Each contest has its own scoring system, so our outcome variable differs by contest; we discuss each choice in turn.

The CF rating is a modified, self-calibrating Elo-rating system inspired by chess ([Mirzayanov 2015](#)). After each contest, a competitor’s rating moves up or down according to how they placed against the others in that round, and the ratings of those opponents enter the calculation directly. The rating therefore already reflects the strength of the field a competitor faced, rather than something we must control for separately. CF also applies an inflation correction that keeps the rating scale roughly stable from year to year, so a rating change of a given size means about the same thing across years. The rating stays relative (a contestant’s standing in the pool, not an absolute skill level), but our CF analyses compare a contestant with their own past and compare subpopulations within the same window, so a common shift in the pool’s level nets out.

In contrast, ICPC and IOI provide no comparable self-calibrating rating. Rather than fall back on the contest rank or medal, a relative outcome that depends on the then-current contestant pool, our primary outcomes are the absolute scores: problems solved at the ICPC World Finals, and total points at the IOI. A score does not depend on the contestant pool at all; it reflects a competitor’s own performance on a fixed problem set. It does, however, vary with the difficulty of each year’s problem set. Accordingly, we include contest-year fixed effects to absorb this variation.

4.4 Descriptive statistics

Table 5 reports descriptive statistics for the variables that enter the three sets of analyses of Section 5, organized so that the panels exhibit the variations that identify the key coefficients of interest. Panel A reports the three AI-prompt practice proxies by cohort and pre/post-ChatGPT period: the

Table 3: Summary statistics for the three analytical panels and the ICPC/IOI–CF linkage. Panels A and B are the ICPC institution-year and IOI country-year panels used for the non-AI-aided-performance tests; Panel C is the combined CF within-individual panel (contest-record roster-linked users and the non-rostered expansion); Panel D summarizes the linkage connecting them. Definitions are given in the table note.

<i>Panel A. ICPC university-year panel</i>	
University-year cells	1,055
Distinct universities	394
spanning pre-AI and AI-era cohorts	112
Cohort: pre-AI / transition / AI-era	272 / 249 / 534
Team CF rating: mean / median / SD	2,150 / 2,146 / 334
Teams with all members CF-linked	768 (72.8%)
Universities by contest year, 2018–2025	138, 134, 117, 132, 124, 130, 141, 139
<i>Panel B. IOI country-year panel</i>	
Country-year cells	708
Distinct countries	103
spanning pre-AI and AI-era cohorts	86
Cohort: pre-AI / transition / AI-era	353 / 89 / 266
Delegation CF rating: mean / median / SD	1,953 / 1,999 / 401
Delegations with all contestants CF-linked	251 (35.5%)
Country-years by contest year, 2018–2025	88, 88, 88, 89, 89, 88, 93, 85
<i>Panel C. CF within-individual panel</i>	
Unique users	10,419
of which spanning subset (≥ 30 subs both windows)	6,399 (61.4%)
User-month observations	793,920
Cohort: pre-AI / transition / AI-era	3,347 / 5,293 / 1,779
spanning subset, same order	2,061 / 4,006 / 332
Cohort CF rating (mean): pre-AI / trans / AI-era	1,818 / 1,610 / 1,531
Practice submissions per user (median)	1,489
Rated contests per user (median)	54
Source: roster-linked / non-rostered	2,480 / 7,939
<i>Panel D. ICPC/IOI $\rightarrow$ CF linkage</i>	
ICPC published-username linkage: accounts	2,189
IOI name-and-country linkage: accounts	1,164
Roster-linked CF users (ICPC $\cup$ IOI), all cohorts	3,117 (236 in both)

Source: authors' calculations over the cleaned panels. Cohorts are assigned by first rated contest date: pre-AI (2015–2018), transition (2019 to the day before the Copilot release), and AI-era (Copilot release to 2024-12-31); users first rated outside 2015–2024 fall out of window and are excluded (Section 4.2). “Spanning” CF users have ≥ 30 practice submissions in both the pre-Copilot and post-ChatGPT windows; spanning universities and countries appear in both a pre-AI and an AI-era cohort cell. Cohort CF rating means are computed over all user-months within each cohort; practice-submission and rated-contest counts are medians over the 10,419-user analysis sample. The published-username linkage uses the CF username that the Hall of Fame lists on each ICPC contestant profile; Panel A and the within-ICPC test (Section 5.3) rely on it, while the name-and-country audit is the linkage route for IOI. Appendix A documents the candidate-matching confidence tiers and the sensitivity of the cross-subpopulation asymmetry to the 1,374 excluded name-only matches. Per-analysis estimation subsamples are documented in Section 4.2.

Table 4: Cohort composition of the CF headline cohort $\times$ post DiD sample (Section 5.1). All counts are over the three cohorts (pre_AI, transition, AI_era). Spanning N marks numbers of users with ≥ 30 practice submissions in both the pre-Copilot and post-ChatGPT windows. Per-user medians are computed on the practice-pattern panel after the 2018–2025 and non-2022-Q4 filters.

Cohort	N	Spanning N	Median rating		Median practice sub counts	
			pre-ChatGPT	post-ChatGPT	pre-ChatGPT	post-ChatGPT
pre_AI	3,347	2,061	1,874	1,874	422	49
transition	5,293	4,006	1,582	1,665	462	142
AI_era	1,779	332	1,348 [†]	1,674	163	408

[†] Median over the 1,336 AI_era users with at least one pre-ChatGPT month of rated activity in the monthly rating panel; the remaining 443 AI_era users entered their first rated contest after ChatGPT and have no pre-ChatGPT rating to summarize. Cohort sample sizes are unbalanced: transition is the largest cohort (5,293) and AI_era is the smallest (1,779), and the spanning sub-sample collapses AI_era to 332 because most AI_era users cannot yet supply 30 pre-Copilot submissions. The DiD uses the full analysis sample (10,419 users; see Section 4.2), not the spanning sub-sample. Median rating is the per-user most recent rating in each window; median practice submissions are per-user totals on the practice-mode panel. Pre/post practice activity moves in opposite directions across cohorts (pre_AI’s median drops from 422 to 49 as veterans exit; AI_era’s median rises from 163 to 408 as new entrants accumulate post-ChatGPT activity); the $\log(1 + n_{h,t})$ control in the DiD absorbs this directly at the user-quarter level.

gap between the AI_era cohort’s pre-to-post change and the pre_AI cohort’s is the raw difference-in-differences that Equation 1 estimates. Panel B reports the cross-subpopulation variables by subpopulation; the within-subpopulation spread of the AI-signature (its 10th-to-90th-percentile range) is the variation that identifies the slopes θ_s , and the level gap between the two subpopulations in baseline rating motivates the $\bar{r}_h^{\text{pre}}$ control. Panel C reports the within-ICPC variables by cohort; the object of interest ψ_{AI} is the difference across cohorts in how AI-intensity relates to score. Figure 1 complements Panel A by tracing the three practice proxies by cohort over the full sample window. The large confidence intervals in the three proxies make it difficult to visually identify a clear pattern change relative to the AI releases; accordingly, the changes are formally identified by the controlled difference-in-differences (Table 6) rather than by the raw cohort levels.

Table 5: Descriptive statistics for the Section 5 analyses. Each panel reports the variables of one analysis, sliced along the dimension that identifies its key coefficient: Panel A by cohort $\times$ pre/post-ChatGPT period (the difference-in-differences of Section 5.1), Panel B by subpopulation (the cross-subpopulation contrast of Section 5.2), and Panel C by cohort within the ICPC sample (the cohort contrast of Section 5.3). Cells are mean (standard deviation) unless noted.

<i>Panel A. CF practice proxies by cohort $\times$ period (Section 5.1; user-quarter panel)</i>					
Cohort, period	Single-shot accept rate	Median attempts	Quick-retry fraction	Subs. / quarter	N user-qtrs
pre_AI, pre	0.429 (0.269)	1.85 (1.59)	0.264 (0.193)	61.6 (94.2)	38,627
pre_AI, post	0.450 (0.325)	1.95 (2.07)	0.228 (0.216)	33.4 (67.7)	13,387
transition, pre	0.449 (0.248)	1.73 (1.34)	0.259 (0.178)	88.6 (129.3)	43,360
transition, post	0.461 (0.290)	1.81 (1.78)	0.247 (0.203)	51.9 (87.5)	32,751
AI_era, pre	0.414 (0.270)	1.87 (1.45)	0.264 (0.191)	78.6 (129.4)	6,558
AI_era, post	0.456 (0.251)	1.74 (1.43)	0.272 (0.188)	79.4 (112.2)	14,780

<i>Panel B. Cross-subpopulation rating asymmetry (Section 5.2; $N = 6,690$ users)</i>			
	Non-rostered ($n = 5,283$)	Roster-linked ($n = 1,407$)	All ($n = 6,690$)
Rating change Δr_h (post – pre)	286.9 (272.4)	350.2 (288.9)	300.2 (277.1)
AI-signature (post-ChatGPT PC1)	–0.003 (1.447)	–0.270 (1.567)	–0.059 (1.477)
p10 / p50 / p90	–1.72 / 0.11 / 1.67	–2.03 / –0.17 / 1.43	–1.79 / 0.05 / 1.62
Baseline rating $\bar{r}_h^{\text{pre}}$	1377 (381)	1766 (451)	1459 (427)

<i>Panel C. Within-ICPC AI-intensity by cohort (Section 5.3; 579 obs, 444 users)</i>			
	pre_AI	transition	AI_era
AI-intensity (first-difference PC1)	–0.150 (1.324)	–0.014 (1.868)	0.161 (0.825)
p10 / p50 / p90	–1.55 / –0.09 / 1.28	–1.26 / 0.12 / 1.41	–0.91 / 0.19 / 1.10
Baseline rating $\bar{r}_h^{\text{pre}}$	2268 (372)	1946 (371)	1338 (337)
ICPC score (problems solved)	5.9 (1.9)	5.1 (2.0)	4.8 (1.5)
N observations	190	309	80
N users	144	233	67

Notes: Cohorts are defined in Table 2. Panel A cells are unadjusted cohort $\times$ period means over user-quarters; the estimated DiD (Table 6) adds user and quarter fixed effects and the $\log(1 + n_{h,t})$ submission control. “pre” is 2018-Q1 through 2022-Q3 and “post” is 2023-Q1 through 2025-Q4 (the 2022-Q4 boundary quarter is dropped); N counts user-quarters in each cell on the single-shot support, and the estimation sample sizes (marginally smaller once fixed-effect singletons are dropped) are reported in Table 6. Panel B reports the cross-subpopulation sample of Section 5.2; “AI-signature” is `ai_pc1_post`, the post-ChatGPT first principal component of the three sign-aligned proxies. Panel C reports the within-ICPC sample of Section 5.3; “AI-intensity” is the within-user first-difference principal component `ai_intensity_pc1`. The IOI sister sample (Section 5.3; 345 obs, 243 contestants) has cohort AI-intensity means of 0.16, –0.24, and –0.14 and IOI-score means of 263, 272, and 227 for pre_AI, transition, and AI_era respectively.

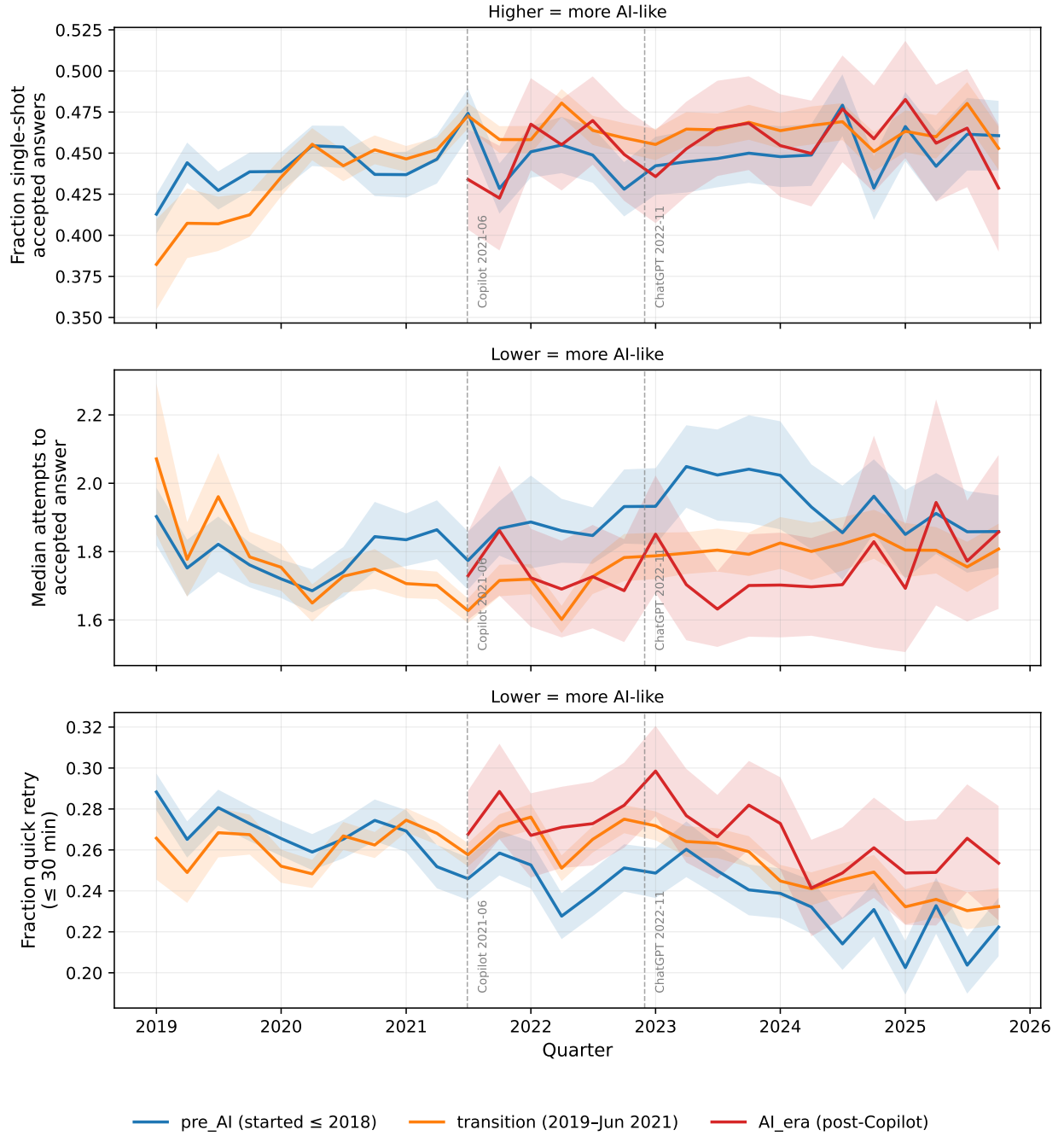


Figure 1: AI-prompt signature components by cohort and quarter. Each panel plots the cohort mean of one practice proxy by quarter with 95% confidence bands; the vertical axis names the metric and the panel title gives its AI-prompt direction. Vertical dashed lines mark the Copilot (2021-06) and ChatGPT (2022-11) releases, and the AI_era series begins after the Copilot launch by cohort construction. To hold the sample fixed across the time series, the figure uses spanning users (those with at least 30 practice submissions in both the pre-Copilot and post-ChatGPT windows; Table 3) in the three focal cohorts.

5 Estimation and Results

This section presents three sets of analyses that jointly distinguish the substitute and complementary roles of generative AI in skill formation; they test Predictions 1–3 of Section 3.3, respectively. The first set (Section 5.1) documents a cross-cohort shift in CF practice patterns toward an AI-prompt signature, larger for cohorts more of whose training postdates AI; we read this shift as descriptive context, because the cohort contrast is not cleanly timed to the AI releases. The second (Section 5.2) shows a cross-subpopulation asymmetry in CF users’ ratings: this same AI-prompt signature predicts rating loss among users in the open CF user pool, but not among rostered users who are linked to AI-prohibited contests, ICPC and IOI. The third (Section 5.3) moves inside the AI-prohibited ICPC environment and finds that AI-intensive CF practice predicts *higher* non-AI-aided scores for AI-era entrants, resolving the asymmetry puzzle: the team formation and qualifier screen substitute-users out, leaving a pool in which the remaining AI use reflects AI-complement practice that enhances frontier skill formation. Section 5.4 discusses one robustness consideration, COVID-19’s impact on the results, with the full battery in Appendix A.

All three analyses share the cohort structure of Section 4.2 (Table 2), with the pre_AI cohort as the reference. The cohort cutoff is anchored at the Copilot launch and the within-user pre/post comparisons at the later ChatGPT release, for the reasons set out in Section 5.1.

5.1 The behavior shift toward an AI-prompt signature in CF practice

We first evaluate how CF practice behavior shifts after the AI releases. Specifically, we focus on three patterns that together form an AI-prompt signature:

1. `frac_single_shot`: the fraction of problems solved on the first attempt;
2. `median_attempts`: the median number of attempts to solve a problem;
3. `frac_quick_retry`: the fraction of quick retries after a failed submission, where a quick retry is defined as a reattempt within thirty minutes of the incorrect submission.

Solving a problem without AI is iterative: the user submits, reads the verdict, locates the bug, and resubmits, often several times before the solution is accepted as correct (“AC,” the platform’s Accepted verdict). A user who relies on AI-assistance can instead submit a generated solution that often clears the problem on the first try, collapsing the submit-debug-resubmit cycle into a single submission, consistent with the keystroke-level workflow shift documented by [Shihab et al. \(2025\)](#). Consequently, if users use AI during practice, three patterns are likely to surface:

1. more problems solved on the first attempt (greater `frac_single_shot`);

2. fewer total submissions per solved problem (lower `median_attempts`); and
3. fewer quick retries after a failed submission (lower `frac_quick_retry`).

Fewer quick retries are the subtlest of the three. Quick retries signal rapid human debugging, where the programmer submits, identifies the bug in code they wrote, and resubmits within a short time span. AI reliance reduces quick retries on two fronts: more first submissions succeed and leave nothing to retry, and, more importantly, when an AI-generated solution does fail, the user often cannot quickly debug code that they did not write themselves.

Throughout, this signature is a behavioral proxy inferred from submission timing. It is consistent with AI-style practice, but not a direct, verified measure of AI use. Appendix B reports construct-validation tests that bound the leading alternative: the signature merely proxies programmer skill because more skilled programmers may show similar behavior patterns even without AI. Specifically, we show that, before AI was available, higher-rated (more skilled) users were, if anything, *less* AI-like on these behaviors, not more (the correlation between CF rating and each signature component is weak, $|r| < 0.20$, and runs opposite to the AI direction), so skill does not produce the signature.

To assess shifts in the three behavioral patterns after AI releases, we consider a difference-in-differences (DiD) design on the user-quarter CF practice panel. For user h in quarter t :

$$y_{h,t} = \alpha_h + \tau_t + \beta_1 \cdot \text{post}_t \cdot \mathbb{1}[\text{cohort}_h = \text{transition}] + \beta_2 \cdot \text{post}_t \cdot \mathbb{1}[\text{cohort}_h = \text{AI_era}] + \lambda \cdot \log(1 + n_{h,t}) + \varepsilon_{h,t}, \quad (1)$$

where $y_{h,t}$ is one of the three patterns, $\text{post}_t = \mathbb{1}[t \geq 2023\text{-Q1}]$ is the post-ChatGPT indicator (the boundary quarter 2022-Q4 is dropped from the panel), α_h are user fixed effects, τ_t are quarter fixed effects, $n_{h,t}$ is the user’s submission count in the quarter, and standard errors are clustered at the user level. The coefficients of interest are β_1 on the transition cohort and β_2 on the AI_era cohort, both relative to the pre_AI reference cohort. The post main effect is absorbed by the quarter fixed effects τ_t .

The sample is the 10,419-user CF panel of Section 4.2, restricted to user-quarter observations on practice-mode submissions from 2018-Q1 through 2025-Q4 (with 2022-Q4 dropped as the ChatGPT boundary quarter): up to 149,398 user-quarter observations, with the exact support varying by outcome (see Table 6). Practice submissions isolate self-directed study from rated-contest activity. Per-cohort composition is in Table 4.

The two AI events anchor this design in different ways. The Copilot launch on 2021-06-29 defines the cross-section: a user’s cohort assignment is set by where their first rated contest falls relative to this date. The ChatGPT release on 2022-11-30 defines the time series: the within-user pre/post indicator post_t flips from zero to one at this date. The two boundaries do not coincide for the following reasons. If both anchors were pinned to Copilot, AI_era users’ pre period would

consist only of whatever practice they did before their first rated contest—a period that spans the novice-to-competitor transition for this cohort. A within-user pre/post comparison anchored at Copilot would therefore conflate that career-stage transition with pre/post AI exposure.

Pinning the time series to ChatGPT instead gives every AI_era user a clean within-user pre period of roughly five quarters (2021-Q3 through 2022-Q3), all observed after the user has entered rated competition, while the cohort cutoff at Copilot still anchors the cross-section. The choice is also substantively defensible. ChatGPT was the broader shock: free, web-based, and immediately mainstream, while Copilot in 2021 was a programmer-specific tool with a much narrower user base. The practice-pattern signature shift visible in the data is also larger around the ChatGPT date than around the Copilot date. A Copilot-anchored DiD on the pre_AI and transition cohorts (AI_era users are unidentified for this contrast by construction) confirms that the practice-pattern shift is detectable around the first AI shock as well, so the headline result is not a ChatGPT-specific artifact (Appendix A.2).

The identifying variation is within-user pre-to-post change in the outcome, differenced across cohorts. User fixed effects absorb time-invariant user characteristics (e.g., skill baseline, education, country); quarter fixed effects absorb calendar-time shocks common to all users, such as platform policy changes and seasonal effects. The specification does not identify the causal effect of AI use. Because cohort assignment is based on platform-entry timing rather than observed AI use, β_1 and β_2 measure the differential pre-to-post change in practice behavior of the transition and AI_era cohorts, respectively, relative to the pre_AI reference. We read these coefficients as a descriptive cohort contrast rather than as an AI effect: as the event study below shows, the gradient across cohorts (β_1 and β_2) does not coincide with the AI releases, so attributing it to AI exposure during training, rather than to a secular change in practice norms or to novice-to-competitor maturation, is not warranted by this design.

Results. The AI_era cohort $\times$ post interaction (β_2) is significant on all three primary outcomes (Table 6). On the single-shot accept rate, $\beta_2 = 0.0586$ (SE 0.0063, $p < 0.001$): in comparison to users with first-rated-contest before 2019 (the pre_AI cohort), AI_era users on average have 5.9 percentage points more problems solved in a single-shot after ChatGPT’s release. On the median attempts to acceptance, $\beta_2 = -0.3271$ (SE 0.0374, $p < 0.001$): AI_era users have roughly a third fewer tries per solved problem. On the quick-retry fraction, $\beta_2 = -0.0190$ (SE 0.0041, $p < 0.001$): AI_era users’ quick-retry fraction falls by 1.9 percentage points. In short, AI_era users’ practice-pattern shifts toward an AI-prompt signature across all three margins.

The transition cohort lies between, but closer to the pre_AI cohort: $\beta_1 = 0.0139$ (SE 0.0046, $p = 0.003$) on single-shot submission and -0.0680 (SE 0.0297, $p = 0.022$) on median-attempts per problem, roughly a quarter of the corresponding AI_era effect. On these two outcomes the

Table 6: AI-Prompt Signature Shifts in CF Practice Patterns (Equation 1)*

Dependent variable	(1) frac_single_shot	(2) median_attempts	(3) frac_quick_retry
post $\times$ transition (β_1)	0.0139** (0.0046)	-0.0680* (0.0297)	0.0090** (0.0029)
post $\times$ AI_era (β_2)	0.0586*** (0.0063)	-0.3271*** (0.0374)	-0.0190*** (0.0041)
$\log(1 + n_{h,t})$ (λ)	-0.0045*** (0.0007)	0.0120** (0.0037)	0.0587*** (0.0005)
User FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Observations**	149,398	142,774	149,441
Users	10,419	10,401	10,419
Quarters	31	31	31

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Notes: *Sample is the CF user-quarter panel, 2018-Q1 through 2025-Q4 excluding 2022-Q4. Reference cohort: pre_AI; the post main effect is absorbed by quarter fixed effects. Standard errors clustered at the user level in parentheses.

**Sample sizes and user counts differ across columns because the three outcomes have different definitional supports: median_attempts is undefined for user-quarters with no solved problems (dropping ~6,600 user-quarters and 18 users); the small gap between frac_single_shot and frac_quick_retry reflects different denominators (problems attempted vs total submissions). Each regression uses the natural sample for its outcome.

gradient pre_AI (reference, 0) $\rightarrow$ transition $\rightarrow$ AI_era is monotonic, consistent with the practice signature strengthening with how much of a user's training occurred post-AI. The quick-retry margin is the exception: the transition coefficient is small and runs the opposite way (0.0090, SE 0.0029, $p = 0.002$) while the AI_era coefficient is negative, so the monotone gradient holds on the other two outcomes but not on quick-retry. One plausible explanation: transition users adopted AI assistance post-shock but retained the iterative debugging habits they formed before AI. Generating code with AI and then debugging by hand produces *more* quick retries than the pre_AI baseline (who use AI less overall) and more than the AI_era cohort, who lack the pre-AI debugging fluency needed to repair AI-generated bugs. On this conjecture, the positive transition coefficient on quick-retry reflects a hybrid complement-style workflow that is masked on single-shot and median-attempts and surfaces here.

This cross-cohort shift is descriptive, and it is not confined to one segment of the platform: re-estimating the frac_single_shot DiD separately by subpopulation, the AI_era coefficient is 0.0356 ($p = 0.004$) among the ICPC-IOI-rostered users and 0.0617 ($p < 0.001$) among the non-rostered users, so the signature is present across the full CF population rather than driven by either pool alone.

The cohort contrast is not cleanly timed to AI. We do not read the cohort gradient (β_1 or β_2) as a causally identified AI effect, for three reasons. First, the parallel-trends assumption behind the DiD fails. We re-estimate Equation 1 with the $\text{post} \times \text{cohort}$ interactions replaced by a full set of $\text{quarter} \times \text{cohort}$ interactions (each quarter dummy interacted with the transition and AI_era indicators, with the pre_AI cohort and the 2022-Q3 quarter as the omitted references), and plot the coefficients in Figure 2: neither the transition nor the AI_era cohort moves discontinuously at the ChatGPT release, and for the transition cohort (the testable contrast, since AI_era users barely appear in the panel before 2023) a joint test of the pre-release coefficients rejects parallel pre-trends on all three practice patterns ($\chi^2(18) = 96.7, 76.8, \text{ and } 84.6$; all $p < 0.001$). Second, the cohort differences do not line up with AI exposure. If AI were the driver, the heavily exposed AI_era cohort should look more AI-like than the transition cohort; instead, at matched career stage (cohorts aligned by quarters since first rated contest, with career-stage fixed effects absorbing the maturation trajectory) the two cohorts are statistically indistinguishable (single-shot difference 0.0002, $p = 0.95$). The gap is between the pre_AI cohort and the later cohorts and does not widen from transition to AI_era, despite AI_era’s far greater AI exposure. Moreover, the transition cohort entered before the Copilot launch, predating both AI releases. The difference is thus a secular, cohort-level shift in practice norms, not one that grows with AI exposure or begins at the AI rollout. Third, the cohort difference in the signature is largest on the problems AI handles worst, not the ones it handles best. Because the signature is first-attempt success, and AI is more likely to supply a correct first attempt on problems it can solve (easy, templated ones, not hard ones), a cohort leaning on AI should show the signature concentrated on easy problems. Instead, holding career stage fixed and splitting by Codeforces problem-difficulty rating, the AI_era-versus-pre_AI single-shot difference is null on the easiest problems ($+0.002$, $p = 0.63$) and rises with difficulty, peaking at $+0.039$ ($p < 0.001$) on the 2000–2399 band before attenuating slightly on the hardest problems. A secular shift in practice norms together with novice-to-competitor maturation fits these three facts better than a clean AI shock, so we carry the practice shift forward as descriptive context that AI alone does not explain. Full event-study, career-time, and difficulty-gradient specifications are available from the author.

Whether AI-style practice translates to frontier-skill erosion at the AI-prohibited contests is the question that the cross-subpopulation asymmetry on CF rating (Section 5.2) probes, and that the within-ICPC result (Section 5.3) tests directly in the AI-prohibited ICPC environment.

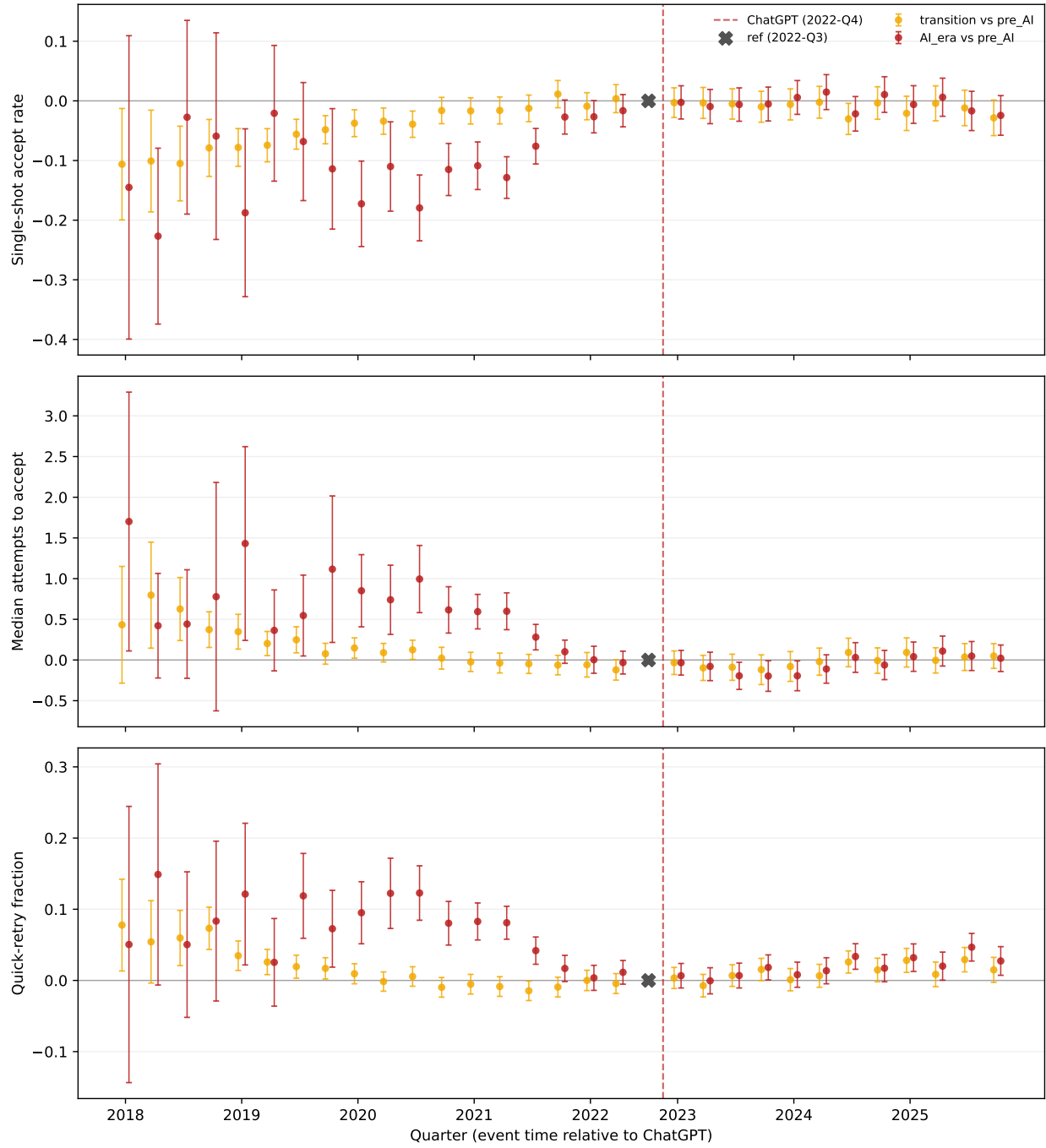


Figure 2: Cohort event study of the three AI-prompt signature proxies (Equation 1 with quarterly event-time interactions). Each panel plots event-time coefficients for the transition (orange) and AI_era (red) cohorts relative to the pre_AI cohort; the reference period is 2022-Q3 (gray marker) and the dashed line marks the ChatGPT release. Bands are 95% confidence intervals (user-clustered standard errors). Neither cohort jumps at the release; AI_era’s pre-2023 leads are uninformative (few users), so the testable pre-trend contrast is the transition cohort, which rejects parallel trends on all three proxies.

5.2 A Cross-subpopulation CF Rating Asymmetry

Next, we assess CF rating changes under the AI-prompt shift discussed above. Specifically, we compare two subpopulations of CF users: roster-linked users who participate in AI-prohibited ICPC and IOI contests, and non-rostered users who do not. The substitute-vs-complement mechanism predicts an asymmetry across these pools rather than a uniform effect: the AI-prohibited contests filter substitute-users out of the roster-linked pool, since their non-AI-aided skill is too weak to qualify. The AI signature should therefore predict rating loss in the non-rostered pool (where substitute-users remain) but be neutralized in the roster-linked pool (where they have been screened out).

We implement this analysis both ways, two subpopulation regressions and one pooled regression.

- The subpopulation version fits two parallel cross-sectional regressions, one per subpopulation $s \in \{\text{non-rostered}, \text{rostered}\}$:

$$\Delta r_h = \alpha_s + \theta_s \cdot \text{ai_pc1_post}_h + \gamma_s \cdot \bar{r}_h^{\text{pre}} + \varepsilon_h, \quad (2)$$

with robust standard errors. The outcome Δr_h is the user’s mean CF rating after ChatGPT minus their mean rating before, computed from the monthly rating panel as $\bar{r}_h^{\text{post}} - \bar{r}_h^{\text{pre}}$, where the post-ChatGPT window covers months $t \in [2022-12, 2025-12]$ and the pre window covers $t \in [2018-01, 2022-12]$.⁴ The covariate $\bar{r}_h^{\text{pre}}$ controls for the user’s baseline skill level.

The coefficient θ_s of `ai_pc1_post` is the coefficient of interest, measuring how AI-signature predicts the subpopulation’s rating change. The covariate `ai_pc1_post` is the user’s post-ChatGPT AI-signature score, defined as the first principal component of the three sign-aligned behavioral AI proxies introduced in Section 5.1 (`frac_single_shot`, the negation of `median_attempts`, and the negation of `frac_quick_retry`). Accordingly, all three components contribute positively to the AI-prompt direction. The bootstrap-validated principal component analysis (PCA) loadings of the three AI-prompt patterns are 0.602, 0.570, and 0.560 respectively.⁵

- The pooled fully-stratified version runs both subpopulations through a single regression with

⁴ChatGPT released on 2022-11-30, so December 2022 falls in the post window. These monthly rating windows split at the release month rather than excising a buffer; this differs from the quarterly Pattern-A difference-in-differences, which drops the 2022-Q4 boundary quarter because its coarser quarterly resolution cannot separate the pre- and post-release weeks within that quarter.

⁵Because the three proxies share a single submission stream, we check that the composite is not one timing pattern restated three times. After sign alignment, the proxies are positively but only moderately correlated (post-shock levels, $n = 8,597$: pairwise Pearson r between 0.45 and 0.68, mean 0.56), so each pair shares roughly a third of its variance, while the first principal component explains 71% of the total. The composite thus aggregates three related but non-redundant indicators rather than a single repackaged variable.

an indicator NR_h for non-rostered users:

$$\begin{aligned} \Delta r_h = & \alpha_0 + \alpha_1 \cdot NR_h + \theta_0 \cdot ai_pc1_post_h + \theta_1 \cdot ai_pc1_post_h \cdot NR_h \\ & + \gamma_0 \cdot \bar{r}_h^{pre} + \gamma_1 \cdot \bar{r}_h^{pre} \cdot NR_h + \varepsilon_h. \end{aligned} \quad (3)$$

The interaction term coefficient θ_1 is the coefficient of interest, measuring the cross-subpopulation asymmetry in rating changes.

The sample consists of users who remain active in the post-ChatGPT window: a user enters if they have at least two quarters between 2023-Q1 and 2025-Q4 with at least five practice submissions in each. After dropping users with missing Δr_h or $\bar{r}_h^{pre}$, the analysis frame contains 6,690 users (5,283 non-rostered and 1,407 roster-linked).

Identification works cross-user within a subpopulation for θ_s (Equation 2) and cross-subpopulation for θ_1 (Equation 3). The within-subpopulation comparison holds the screening environment fixed and asks how AI-style behavior co-moves with rating change among users facing the same gate (or no gate). The cross-subpopulation comparison holds the AI signature fixed and varies the screening environment. The asymmetry test relies on the testable assumption that the roster-linked pool has been screened of substitute-users by the AI-prohibited qualifying rounds; so it does not require roster status to be randomly assigned.

The inclusion of baseline-skill control $\bar{r}_h^{pre}$ is crucial because it blocks omitted-variable bias. The AI signature ai_pc1_post correlates negatively with $\bar{r}_h^{pre}$, since high-AI users are on average lower-skill, and $\bar{r}_h^{pre}$ independently predicts Δr_h (its coefficient is -0.32 in non-rostered and -0.34 in roster-linked, both with $p < 0.0001$). Without the control, the AI coefficient would be biased.

We want to highlight that the specification does not identify the *causal* effect of AI use on rating change. The AI signature is observational, and high-signature users may differ from low-signature users on unobservables. We control for baseline skill and exploit the cross-subpopulation screening difference, not random assignment of AI use; the asymmetry test is interpretable as evidence of a screening-filter under the maintained assumption that the ICPC-IOI qualifier screens substitute-users out of the roster-linked pool.

Results. The within-subpopulation slopes on ai_pc1_post are starkly asymmetric (Table 7). The non-rostered slope is -12.24 rating points (SE 2.45, $p < 0.0001$, 95% CI $[-17.04, -7.44]$). The roster-linked slope is statistically null: -2.41 (SE 4.33, $p = 0.58$). The interaction coefficient θ_1 of the pooled regression is -9.82 (SE 4.98, $p = 0.048$, 95% CI $[-19.58, -0.07]$): the AI signature predicts a rating loss around ten points larger in the un-screened pool than in the screened roster-linked pool.

The asymmetry is in the direction the screening-filter mechanism predicts. The AI signature

Table 7: Cross-subpopulation CF rating asymmetry on `ai_pc1_post`. Sample: $N = 6,690$ users active in the post-ChatGPT window (5,283 non-rostered + 1,407 roster-linked). Robust standard errors. The fully-stratified pooled spec tests $\theta_{\text{non_rostered}}$ and θ_{rostered} being statistically different.

Coefficient	θ	SE	p	95% CI
<i>Subpopulation specifications (Equation 2)</i>				
Non-rostered slope $\theta_{\text{non_rostered}}$	-12.24	2.45	< 0.0001	[-17.04, -7.44]
Roster-linked slope θ_{rostered}	-2.41	4.33	0.58	[-10.91, 6.08]
<i>Pooled fully-stratified spec (Equation 3)</i>				
<code>ai_pc1_post</code> × <code>non_rostered</code> (θ_1)	-9.82	4.98	0.048	[-19.58, -0.07]

predicts rating loss in the non-rostered pool, where substitute-users still appear. The same signature is uncorrelated with rating change in the roster-linked pool, consistent with the AI-prohibited ICPC and IOI qualifiers having filtered substitute-users out. This roster-linked null is a failure to reject rather than a precise zero (95% confidence interval [-10.91, 6.08]), and the differential-attrition check below bounds how much of it could reflect sample selection rather than screening.

However, the asymmetry on CF rating, an outcome where in-contest AI use is hard to police, raises a reasonable objection: if roster-linked users use AI more proficiently during CF contests than non-rostered users, the gap could be measuring who uses AI less effectively during CF contests rather than who developed weaker non-AI-aided skill. The within-ICPC result (Section 5.3) addresses this objection by moving to the AI-prohibited ICPC environment where in-contest AI use is procedurally blocked. Together with the positive within-ICPC slope (Section 5.3), this negative slope yields the same-training-inputs, opposite-signs pattern that operationally distinguishes substitute use from complement use of AI.

Selection robustness. The post-ChatGPT activity requirement (≥ 2 quarters with ≥ 5 practice submissions each) is needed for `ai_pc1_post` to be reliably measured at the user level, but it cuts the sample from 10,419 users (the Section 5.1 sample) to 6,690, raising a sample-selection concern that we address in three ways (Appendix A.3 reports the full results). First, the cross-subpopulation interaction θ_1 is stable under filter relaxation: weaker filter thresholds (fewer required quarters and submissions) yield -11.92 ($p = 0.017$) and -7.14 ($p = 0.081$), bracketing the headline -9.82. Second, the Section 5.1 practice-pattern shift survives on the Section 5.2 sub-sample with 8 to 13% attenuation across all three outcomes (all $p < 0.0001$), confirming that the same population drives both findings. Third, roster-linked users drop out of the activity filter more often than non-rostered users (43.3% vs 33.5%), likely because ICPC and IOI contestants step back from open CF practice after their qualifying year. The remaining roster-linked sample is therefore more selected than the non-rostered sample, and the asymmetry magnitude varies non-monotonically across filter thresholds (the [-7.14, -11.92] range above brackets -9.82). Appendix A.3 reports the full balance

decomposition by subpopulation.

Selection bounds. We bound the cross-subpopulation interaction θ_1 against selection on unobservables and against the differential roster-linked attrition just noted (Table 8). Against selection that mimics the observables, the asymmetry is robust: adding the baseline-rating control barely moves θ_1 (from -10.04 to -9.82), so by the coefficient-stability logic of [Oster \(2019\)](#) an unobservable would have to be roughly twelve times as important as all observables combined to drive θ_1 to zero, and the asymmetry holds within baseline-rating-by-cohort matched cells (coarsened exact matching: $\theta_1 = -9.73$, $p = 0.050$). It is not, however, robust to the worst case of the differential attrition between the two subpopulations. Since the non-rostered pool is retained more often than the roster-linked pool, the Lee trimming bound ([Lee 2009](#)) brackets θ_1 in $[-11.95, -0.35]$: benign attrition (-11.95) leaves the asymmetry intact, while the worst-case trim shrinks it to near zero (-0.35). Relatedly, given the marginal significance of θ_1 ($p = 0.048$), its robustness value ([Cinelli and Hazlett 2020](#)) is small: a confounder explaining only 2.4% of the residual variance in both the treatment and the outcome would be enough to zero θ_1 . As a benchmark, the baseline rating control explains 15.7% of that residual variance, so a confounder far weaker than baseline rating would suffice to overturn the result. We therefore read the cross-subpopulation asymmetry as suggestive and attrition-bounded rather than decisive, and rest the mechanism evidence on the within-ICPC test (Section 5.3), which is measured in an environment where AI use is procedurally blocked and whose positive sign survives its robustness battery, even though its magnitude is construct-dependent.

Table 8: Selection bounds on the cross-subpopulation interaction θ_1 (headline -9.82 , $p = 0.048$). The asymmetry is robust to selection on observables (Oster; matching) but bounded by the differential roster-linked attrition (Lee) and fragile at the significance margin (Cinelli–Hazlett).

Method	Result for θ_1	Reading
Oster (2019) δ for $\theta_1 = 0$	12.6 to 147	Robust to observable-like selection
Coarsened exact matching	-9.73 ($p = 0.050$)	Holds on baseline-rating $\times$ cohort support
Lee (2009) attrition bound	$[-11.95, -0.35]$	Near zero in the worst case
Cinelli and Hazlett (2020) $RV_{q=1}$	0.024 (vs. 0.157, baseline rating)	Fragile at the $p = 0.048$ margin

The Oster δ range spans $R_{\max}^2 = 1$ to $R_{\max}^2 = 1.3\tilde{R}^2$; $\delta > 1$ means unobservable selection must exceed observable selection to overturn the estimate. The Lee bound trims the over-retained non-rostered pool to the roster-linked retention rate. $RV_{q=1}$ is the share of residual variance a confounder must explain in both treatment and outcome to zero the estimate. Full results available from the author.

5.3 AI-complement practice and higher non-AI-aided scores

The third set of analyses asks whether AI-intensity during CF practice predicts AI-prohibited ICPC performance differently across cohorts. We estimate a pooled cross-sectional ordinary least squares (OLS) regression with university fixed effects and team-clustered standard errors:

$$s_{h,y} = \alpha + \psi_0 \cdot \text{ai_intensity_pc1}_h + \sum_{c \in \{\text{trans}, \text{AI}\}} \psi_c \cdot \text{ai_intensity_pc1}_h \cdot \mathbb{1}[\text{cohort}_h = c] + \gamma \cdot \bar{r}_h^{\text{pre}} + \delta_y + \mu_u + \varepsilon_{h,y}, \quad (4)$$

where each observation is a user h appearing at an ICPC contest in year y , δ_y is a contest-year fixed effect, and μ_u is a university fixed effect.

The outcome $s_{h,y}$ is the ICPC score of user h 's team in year y (the number of problems the team solved at the contest). ICPC is a team competition: three contestants compete as one university team, so the same team score outcome is shared within team. This shared-outcome structure motivates the team-level clustering of standard errors discussed below. The treatment `ai_intensity_pc1` measures each user's within-user shift toward AI-style CF practice. Each of the three AI-prompt proxies of Section 5.1 is computed over a pre window (2021-Q3 through 2022-Q3, the five quarters between Copilot and ChatGPT releases) and a contest-year-conditional post window (2023-Q1 through the quarter immediately preceding the contest); the user-level change on each proxy (the first difference) is the post-window mean minus the pre-window mean. `ai_intensity_pc1` is the first principal component of the three sign-aligned changes. The PCA loadings are fit on a broad CF reference group ($n = 4,959$) and then frozen and applied to the within-ICPC sub-sample.⁶ The PCA loadings on `frac_single_shot`, the negation of `median_attempts`, and the negation of `frac_quick_retry` are 0.630, 0.530, and 0.568 respectively. The first-difference construction nets out each user's pre-shock baseline practice style, so the treatment captures the within-user *change* in signature rather than the cross-user level. We assign each user a single value, anchored at their first qualifying year, so that the treatment is a time-invariant, pre-determined user characteristic rather than one per user-by-contest-year.⁷ This choice is deliberate on two grounds. First, anchoring at the user's entry into the qualifying pool measures the practice shift before any contest outcome could feed

⁶We fit the loadings on a broad reference group rather than on the 444 within-ICPC users so that the axis is estimated from population-level co-variation and is not defined by the same sample on which it is then tested. The reference group is every user in the CF practice panel (all cohorts, not restricted to the ICPC or IOI rosters) with ≥ 3 active quarters in the pre-window (2021-Q3 through 2022-Q3) and ≥ 2 active quarters in the full post-window (2023-Q1 through 2025-Q4), where a quarter is active if it has ≥ 5 practice submissions. About 91% of these users are in the analysis panel; the remainder are out-of-window-cohort or name-only-matched users that the panel excludes. The frozen loadings are then applied to each within-ICPC user's contest-year-conditional first-differences to produce that user's `ai_intensity_pc1`.

⁷The anchor is the user's earliest year in which both the pre- and post-windows contain sufficient practice activity; for users with multiple 2023+ ICPC contests this can differ from the year of the literal earliest contest when that earliest year lacked qualifying activity.

back into how the user practices. Second, it avoids a mechanical artifact: a contest-year-conditional value would drift with the length of the post window—one or two quarters for a 2023 contest versus eight or more for a 2025 contest—rather than with any genuine change in practice. The baseline-skill control $\bar{r}_h^{\text{pre}}$ (defined as in Section 5.2) is included for the same omitted-variable reason. Cohort assignment follows Section 4.2 (Table 2), with pre_AI as the reference. Contest-year fixed effects δ_y absorb year-specific score-distribution variation (different problem sets, different scoring rubrics across years), and university fixed effects μ_u absorb the time-invariant university-selection layer described below.

The combination of university fixed effects and team-clustered standard errors addresses two distinct identification issues. The first is selection into university. Top universities recruit more homogeneous high-skill students, and subsequently their teams may contain both higher scores and more AI-era high-AI-intensity members. With the fixed effects, the AI-era $\times$ AI-intensity coefficient identifies off within-university variation only, comparing team members at the same university across years. The second issue is the shared team score within team members described above: residuals are mechanically correlated within team, and team-level clustering captures that correlation. With university fixed effects already absorbing the time-invariant university-selection component, the residual structure across teams within a university is plausibly small, and team-level clustering is enough. A more conservative alternative would cluster at the university level ($n = 196$); this inflates the standard error on the headline coefficient from 0.087 to 0.104 and shifts the p -value from 0.028 to 0.068, leaving the headline marginal but the point estimate unchanged. Dropping the university fixed effects entirely (and clustering at the team level) returns a larger $\psi_{\text{AI}} = 0.323$ (SE 0.157, $p = 0.040$), because cross-university confounding is no longer absorbed; we therefore treat the within-university, team-clustered version as the headline.

The sample for this analysis is the intersection of the three cohorts and the ICPC roster, restricted to contest years on or after 2023 where AI-intensity is meaningfully defined. Table 9 reports the chain that produces the sample; Appendix A.4 examines how sensitive the result is to the Step-3 activity filter and to the treatment construct.

The coefficient of interest, ψ_{AI} on the interaction between `ai_intensity_pc1` and the AI-era cohort indicator, identifies how the AI-intensity–score relationship differs between the AI-era and pre_AI cohorts (within a university and a contest year, and holding baseline skill fixed), rather than the causal effect of AI use on score. The pre_AI cohort serves as a built-in placebo: its members had little AI exposure during training, so its slope ψ_0 should be near zero under any reading of the mechanism, and the substantive test is whether the AI-era slope is distinguishable from it. The estimate is observational rather than experimental: the within-user change in AI signature is not randomly assigned, so AI-era users who shifted more toward AI-style practice may differ on unobservables from those who shifted less. The cross-cohort comparison conditions on baseline skill

Table 9: Sample filter chain for the within-ICPC regression (Equation 4). The within-ICPC regression and the practice-pattern DiD (Equation 1) share the same analysis sample of 10,419 users; the within-ICPC regression retains 444 users after the ICPC-specific filters, yielding 579 user-by-contest-year observations across 321 teams in 196 universities.

Step	Filter	Users	Drop	Retention
0	The three cohorts in the CF panel (shared with Equation 1)	10,419	—	—
1	Linked to any ICPC team roster entry	1,673	−8,746	16.1%
2	ICPC participation in contest year ≥ 2023	1,026	−647	61.3%
3	<code>ai_intensity_pc1</code> defined (qualifying pre- and post-window practice activity)	447	−579	43.6%
4	$\bar{r}_h^{\text{pre}}$ also defined (rated CF history before ChatGPT)	444	−3	99.3%

and contest-year fixed effects but cannot rule such differences out; we therefore read the coefficient through the screening-filter lens discussed with the results below, not as a causal effect of AI use.

Results. The AI-era cohort interaction is positive and significant: $\psi_{\text{AI}} = 0.190$ (SE 0.087, $p = 0.028$). For pre-AI users (the reference cohort and the de facto placebo), the slope is small and null ($\psi_0 = -0.048$, SE 0.047, $p = 0.303$). For transition users it is also null ($\psi_{\text{trans}} = 0.060$, SE 0.068, $p = 0.373$). The slope on AI-intensity therefore differs across cohorts only for the AI-era group, with the sign in the predicted direction: among AI-era users, who started their competitive programming careers with AI tools, a larger within-user shift toward AI-style practice predicts *higher* ICPC score in 2023–2025, controlling for baseline skill and university (Table 10).

Table 10: Within-ICPC AI-intensity by cohort. Headline specification with university fixed effects and team-clustered standard errors. Sample: 579 user-by-contest-year observations across 444 users, 321 teams, and 196 universities. Contest years 2023–2025. Reference cohort: pre-AI.

Term	ψ	SE	p	95% CI
<code>ai_intensity_pc1</code> (pre-AI reference)	−0.048	0.047	0.303	[−0.140, 0.043]
<code>ai_intensity_pc1:transition</code>	0.060	0.068	0.373	[−0.072, 0.193]
<code>ai_intensity_pc1:AIera</code>	0.190	0.087	0.028	[0.020, 0.360]
$\bar{r}_h^{\text{pre}}$	0.001	0.000	< 0.001	[+0.000, 0.001]
University FE	✓			
Contest-year FE	✓			
SE clustering	team			

This positive sign inside the AI-prohibited environment is the cross-environment counterpart to the cross-subpopulation asymmetry’s negative slope in the un-screened CF pool (Section 5.2). The same training input (a shift toward AI-style practice) maps to opposite signs depending on whether

the contest environment screens substitute-users out. Under the screening-filter reading, the ICPC qualifier has already removed users whose AI-style practice reflects substitution; the within-user shift toward AI among the qualified pool then reflects complement-style adaptation that raises score, not substitution that erodes it.

Construct and selection robustness. Three caveats apply to the headline estimate $\psi_{AI} = 0.190$. First, the construct is a within-user first difference, aggregated to a single per-user value anchored at the user’s first qualifying year, with the principal-component loadings fit on a broad reference group rather than on the within-ICPC sample. We adopt this construct on *ex ante* grounds, before examining the outcome: a within-user skill-formation question calls for the *change* in practice style rather than its level (the first difference nets out each user’s fixed baseline style), anchoring at the first qualifying year measures the shift before any contest outcome can feed back into practice, and fitting the loadings on the broad reference group keeps the axis from being defined by the same sample on which it is tested.⁸ The estimate is nonetheless construct-dependent. Among the four constructs in Appendix Table 17, the AI-era slope is positive in three and significant only under this preferred construct (the per-(user, year) change gives 0.108, $p = 0.611$; levels-based constructs are null or negative), and across activity thresholds it ranges from 0.11 to 0.43, positive throughout. We therefore read the within-ICPC evidence as establishing the *sign* of the effect, the direction that completes the opposite-signs test with Section 5.2, rather than the precise magnitude 0.190. Second, the pre-window of the first-difference (2021-Q3 through 2022-Q3) partly predates the rated competition careers of AI-era users (whose first rated contest is post-Copilot by construction): these contestants are the novices in the 2023–2025 sample (mean tenure 3.9 years since their first rated contest, against 7.4 years for the pre-AI cohort), so the AI-era change could in part capture novice-to-competitor maturation rather than an AI-related shift. To test this, we add exogenous career-stage controls to Equation 4: tenure, tenure interacted with cohort, and cumulative rated-contest experience. The AI-era slope does not move: ψ_{AI} goes from 0.190 to between 0.195 and 0.196 (p between 0.025 and 0.027) across these specifications, and it survives even the over-control of conditioning on within-window rating growth (0.190, $p = 0.038$), itself partly an outcome of AI use. Were the slope elite-novice maturation, holding career stage fixed would attenuate it; it does not, so the novice-growth confound is bounded.⁹ Third, the within-ICPC sample is reached through the filter chain in Table 9, and two steps raise selection concerns: the Step-3 activity filter drops 579 (56%) of ICPC participants who lack the practice activity needed to measure a first-difference, and the Step-2 participation cutoff drops 647 users with pre-2023 ICPC history but no recent participation. The Step-2 exit is ambiguous between graduation, loss of interest, and

⁸A levels-based measure, suited to the cross-sectional comparison of Section 5.2, would instead conflate the change with the level and is reported for completeness in Appendix A.4.

⁹Full specification table available from the author.

AI-driven dropout. The last channel, if present, would select against exactly the substitute-users the screening-filter reading assumes are absent. Appendix A.4 reports the full construct sensitivity, filter sensitivity, balance check, and Section 5.1 cross-check.

IOI sister analysis. The IOI counterpart to the within-ICPC AI-intensity test is weaker than ICPC’s in two ways. First, precision: the IOI is a smaller pool and the test runs on only 345 contestant-year observations from 243 CF-linked contestants over the 2023–2025 window, in comparison with the 579 observations from 444 users for ICPC. So the estimate is far less precise. Second, age and timing: IOI participants are typically high-schoolers aged 17–19, while ICPC participants are college students aged 18–23, so a higher share of an IOI contestant’s skill-formation window is post-AI, leaving less pre-AI training to anchor the cohort contrast.

Nevertheless, we estimate the same AI-intensity-by-cohort specification on the IOI individual contestants linked to CF accounts, clustered at the contestant level. The AI_era cohort interaction on IOI score is 18.1 (SE 18.0, $p = 0.31$): directionally consistent with the ICPC result but far noisier, with a 95% confidence interval spanning zero. We anchor the non-AI-performance headline on ICPC and treat IOI as a directional check rather than a load-bearing test.

5.4 Robustness

COVID disruption (2020–2021). The empirical window straddles the COVID-19 pandemic. ICPC 2020 was postponed and held in 2021 with reduced attendance; IOI 2020 was held virtually with reduced participation; multiple national qualification pipelines were disrupted across 2020 and 2021. The CF practice panel is the most COVID-resistant of the three samples because it is continuous rather than tied to an annual contest schedule; the cohort $\times$ post DiD on practice-pattern proxies (Section 5.1) absorbs year-level shocks through its quarter fixed effects, and within-user rating dynamics include 2020 and 2021 observations as part of the natural trajectory rather than as anomalies to be dummied out. The cross-subpopulation CF rating asymmetry (Section 5.2) is similarly COVID-resistant: the rating-change outcome Δr_h is constructed as a pre-ChatGPT-versus-post-ChatGPT difference where both windows span multiple years, washing out year-specific shocks. The within-ICPC AI-intensity test (Section 5.3) restricts to contest years ≥ 2023 , which entirely post-dates COVID, so the COVID years cannot directly contaminate the AI-intensity coefficient. Where a specification is not structurally immune in this way, namely the event studies, which estimate a coefficient for each calendar year, we make an explicit adjustment rather than rely on the design: we set the reference period to 2019 (the last pre-COVID year) instead of the COVID-adjacent 2022, and absorb 2020 and 2021 with their own dummies.

Broader robustness battery. Appendix A reports the remaining robustness checks: the ICPC/IOI–CF linkage and the sensitivity of the cross-subpopulation asymmetry to the excluded name-only matches, the Copilot-anchored difference-in-differences, and the selection-bias robustness for the cross-subpopulation and within-ICPC tests.

6 Discussion

The findings combine into one internally consistent picture: a cross-cohort shift in practice patterns toward an AI prompt-and-fix style, which we read as descriptive; a cross-subpopulation asymmetry in which the AI-prompt signature predicts CF rating loss in the un-screened, non-rostered pool but not in the screened, roster-linked pool, bounded by differential attrition; and a robust positive within-ICPC association between a shift toward AI-style practice and non-AI-aided contest score among AI-era entrants who survived the qualification gate. These pieces fit together only under the selection-into-AI-prohibited-contests structure, which is the central contribution of this paper.

6.1 Theoretical contributions

This paper makes three theoretical contributions to the literature on AI and skill formation. First, the paper introduces the selection-into-AI-prohibited-contests structure as a load-bearing feature of empirical designs in this literature. Competitive credentialing systems with AI-prohibited gates filter substitute-dependent agents out of the elite pool. This filtering changes which signals appear in which observational windows: the substitution-driven rating signal from substitute-users is observable only on AI-allowed platforms, while AI-prohibited contests, having screened those users out, reveal only the complement-user association among the deep-skill subpopulation that remains. Recognizing this selection structure reconciles the cross-subpopulation and within-ICPC findings that would otherwise appear contradictory, and it has implications for any empirical design that uses credentialing-gate variation to identify AI substitution effects.

This first contribution is, more specifically, a contribution to information systems. The AI-prohibited evaluation gate is a designed IT-enabled institution, not merely an economic screening device: its rules (no AI, in-person, proctored) and its qualification structure are design choices that determine whether differential AI use is rendered visible or screened out. The AI-prompt signature, in turn, is a measure of technology-in-practice (Orlikowski 2000) rather than of technology adoption: it captures how AI is enacted in routine work, not whether a user can access it. Our central result, that the same enacted practice carries opposite performance signals depending on the screening design of the evaluation environment in which skill is later tested, is therefore a statement about sociotechnical evaluation-system design. A purely labor-economics reading would frame the same

evidence as human-capital depreciation; the information-systems reading locates the policy lever in the design of the evaluation artifact, which is where institutions can act.

Second, the paper operationalizes the mechanism of human capital depreciation in [Acemoglu et al. \(2026\)](#) at the individual frontier scale, providing a multi-year, field-observational counterpart to the macro-theoretical argument and the laboratory evidence in [Bastani et al. \(2025\)](#), [Stadler et al. \(2024\)](#), and [Shen and Tamkin \(2026\)](#). The cross-cohort practice-pattern shift we document, toward an AI prompt-and-fix submission style at multi-year scale, is consistent with the debug-skipping mechanism that [Shen and Tamkin \(2026\)](#) identify in an RCT.

Third, the paper raises broader implications for the value of frontier skill in an AI-augmented world. Five considerations support continued investment in non-AI-aided capability even as AI tools become ubiquitous:

1. **AI failure modes and verification.** AI hallucinates, fails on novel problems by design (it is recombination over the training distribution), and produces plausible-looking output that is subtly wrong. Catching AI errors requires non-AI-aided expertise. [Bastani et al. \(2025\)](#) found exactly this: GPT-Base students performed 17% worse on non-AI-aided exams *and did not perceive they were performing worse*. Self-perception is unreliable; a workforce dependent on AI but lacking non-AI-aided skill to verify AI output cannot detect propagating errors.
2. **Frontier creation, not just frontier execution.** AI is a powerful interpolator across its training distribution but a poor extrapolator beyond it. Genuine breakthroughs — novel theorems, new scientific paradigms, out-of-distribution insights — remain primarily human contributions, even as AI accelerates surrounding execution. [Acemoglu et al. \(2026\)](#) formalizes this: if humans stop generating outside-distribution ideas, the training distribution stagnates and AI itself ceases to improve.
3. **Critical windows in skill formation.** [Stadler et al. \(2024\)](#)’s mediation result (germane cognitive load fully mediates AI’s effect on output quality) implies that AI substitutes precisely for the deep cognitive engagement that builds schemas. Schemas built early provide the substrate for later expertise; if high-potential entrants fail to build them during their formative years, they may never build them, even if AI is later removed.
4. **Welfare resilience under AI-access disruption.** AI access can be revoked through subscription cost, political restriction (Italy’s 2023 ChatGPT ban; China’s continual restriction; EU AI Act effects), corporate policy, infrastructure failure, security breach, or geopolitical isolation. Populations dependent on AI lose capability instantly when AI access disappears; populations with non-AI-aided skill retain a fallback.

5. **Verification in trusted institutions.** Courts, peer review, medical diagnosis, scientific reproducibility, and financial auditing rely on humans who can independently verify claims without AI mediation. The verification crisis is downstream of, but distinct from, the production crisis.

6.2 Practical implications and policy levers

The findings suggest two complementary policy levers for preserving non-AI-aided frontier capability in elite skill markets. They operate at different scales and through different mechanisms, and they reinforce rather than substitute for each other.

Lever 1: within-training, complement-style AI integration. Our within-gate result makes this lever more than a conjecture. Among survivors of the qualification gate, a stronger shift toward AI-style practice is associated with *higher* non-AI-aided ICPC performance (Section 5.3): the complement channel, AI that supports rather than supplants skill formation, is visible among elite performers. What our design cannot identify is whether the *mode* of integration can be steered to put more users on that channel; Bastani et al. (2025)’s classroom RCT supplies the causal step, showing that scaffolded AI (problem-specific hints rather than full solutions) eliminates the substitution effect on non-AI-aided performance. Read together, the field pattern and the RCT motivate a training principle for elite settings: integrate AI as a scaffolded complement rather than a full-solution substitute.

Lever 2: AI-prohibited evaluation gates at the population level. The selection-into-AI-prohibited-contests structure functions as a population-level filtering mechanism. Programmers whose competitive performance relies heavily on AI cannot pass AI-prohibited qualification rounds; the elite pipeline retains deep-skill investors while substitute-users remain visible only on AI-allowed platforms. The cross-subpopulation asymmetry (Section 5.2) is consistent with the filter screening substitute-users out before the contest. The within-ICPC test (Section 5.3) is consistent with the same filter: the positive within-gate association noted under Lever 1 reflects complement-user selection, the substitute-users having been screened out upstream. The lever is the existence and design of AI-prohibited evaluation moments at the population scale, not the elimination of AI from training.

Selection vs. anticipation: what we identify and what we do not. The mechanism we identify for Lever 2 is *selection*: AI-prohibited gates filter substitute-dependent programmers out of the elite pipeline. A separate, complementary mechanism — *anticipation* — would operate behaviorally: programmers who know the AI scaffold will be removed at critical evaluation moments invest more in deep skill during training and use AI as a complement rather than a substitute. The two mechanisms have different population-level implications. Under selection alone, AI-prohibited

gates sort but do not change individual behavior; under anticipation, the gates’ existence shifts the training distribution itself. Our data are consistent with both channels but cannot distinguish them. Identifying the anticipation channel would require a natural experiment in gate-introduction (or removal), or a longitudinal design tracking training-behavior changes after credible AI-prohibition signals. We flag this as productive future work and rest the policy argument here on the selection channel that we identify.

This policy argument rests on the screening we document, not on a proven erosion effect. The opposite-signs pattern shows that AI-prohibited gates separate practice types, so the gate is the relevant lever whether or not substitute-style practice ultimately erodes frontier skill; if there is a long-run cost, it concentrates among the substitute-users the gate screens out. The argument is population-level, not within-individual: AI access is privately valuable to most users, and the externality, if any, is at the level of the elite pool from which frontier capability is drawn.

6.3 External validity

The argument for AI-prohibited evaluation gates has natural limits, and the lesson is not “ban AI from all evaluation.” Three considerations bound the generalization from competitive programming to other elite skill markets.

Existing gate infrastructure varies by field. Medicine, law, and scientific peer review already have AI-prohibition structures that approximate our setting. Medical board exams (USMLE Step 1–3, internal medicine recertification) are proctored and AI-prohibited; the bar exam similarly restricts AI use during the timed test. Scientific peer review has informal but increasingly explicit AI-prohibition norms at journals and grant agencies. Within these fields, the selection-into-AI-prohibited-contests framework should extend: substitute-users of AI during training would face the same kind of gate-level filter that competitive programmers face at ICPC qualification. Fields without such gates — much of corporate software engineering, marketing analytics, financial services analysis — do not have a comparable filtering mechanism, and the population-level externality would operate without the screening buffer that our setting documents.

Designing AI-prohibited evaluations is itself difficult. Even where gate structures exist, designing evaluations that effectively prohibit AI use is increasingly hard as AI becomes embedded in evaluation infrastructure (proctoring tools, scoring systems, candidate-side LLM-assisted note-taking on un-proctored portions). Our setting (ICPC, IOI) benefits from on-site proctoring at qualification and final-round stages; many other fields cannot replicate this. The lesson is that AI-prohibited evaluation gates have a filtering function that compounds with within-training safeguards, and removing or weakening them has population-level consequences worth weighing against the productivity and capability gains of AI-permissive evaluation.

Different fields face different AI-erosion risks. The competitive programming setting is one where AI tools are unusually effective at the focal task (code generation, debugging, problem decomposition). Fields where AI is currently weaker — novel mathematical proof, surgical decision-making, qualitative interpretation in social science — face a smaller substitution-erosion risk in the near term, though this is a moving target as model capabilities advance. The framework is robust to the substitution rate in any specific field; what varies is the urgency of the policy lever.

6.4 Non-AI alternative explanations

The alternatives we name below operate across subpopulations, so they bear chiefly on the cross-subpopulation CF rating asymmetry (Prediction 2, Section 5.2). The within-ICPC AI-intensity slope (Prediction 3, Section 5.3) is identified within the screened pool, and the within-user practice-pattern shift (Prediction 1, Section 5.1) within the same users across the AI shock; both are correspondingly less exposed to these channels.

Tech labor market shocks during 2022–2024. Macroeconomic and tech-sector labor-market shocks over this period may have hit countries with the strongest pre-AI competitive-programming bases asymmetrically: elite programmers there may have moved into industry roles or relocated, depleting the AI-prohibited-contest pipeline. The cross-subpopulation asymmetry could partly capture this depletion rather than direct AI-driven skill erosion.

Demographic and curricular shifts. Top universities increasingly redirect elite computer-science talent toward applied AI/ML curricula. If elite institutions in those countries pivoted faster, traditional competitive-programming participation may have declined as a side effect rather than direct AI-driven skill erosion.

Regression to the mean. Year fixed effects absorb the average; the within-subpopulation differential underlying the cross-subpopulation asymmetry is not mechanically driven by mean reversion. But if some subpopulations had unusually strong cohorts going into 2022, modest reversion could amplify the post-2022 differential. The roster-linked subpopulation is defined ex ante by ICPC participation, which makes pure reversion-to-the-mean a plausible alternative to active erosion for the cross-subpopulation specification.

For the practice-pattern shift (Prediction 1) we do not claim AI substitution over these alternatives: we treat the cohort AI-style practice result as descriptive rather than causal. The screening argument rests instead on the *joint* pattern of Predictions 2 and 3, which the alternatives do not naturally produce: labor-market, curricular, or reversion-to-the-mean shifts would not generate a negative cross-subpopulation asymmetry coexisting with a positive within-ICPC slope (non-rostered $\theta_{NR} < 0$, roster-linked $\theta_{RL} \approx 0$, within-ICPC $\psi_{AI} > 0$). The selection-into-AI-prohibited-contests structure organizes that opposite-signs pattern with a single mechanism, whereas the non-AI alternatives would

each have to explain the two signs separately. That said, we cannot fully rule the alternatives out, and the cross-subpopulation asymmetry in particular remains attrition-bounded; our interpretation acknowledges this uncertainty.

6.5 Limitations and future research

Identification limits. Observational identification has bounded reach. Selection-on-entry into the broad CF population is partially handled by within-individual fixed effects, but selection on continued participation, on AI tool use, and on the choice to enter qualification rounds is not fully eliminated. Pre-screening at country-level olympiad selection gates (which themselves prohibit AI) further attenuates the substitution signal in the linked ICPC/IOI cohort relative to the broader competitive-programming population, making contest-level effects a conservative test. The COVID-aware robustness reported in Section 5.4 defends the primary identification but does not resolve all identification concerns.

Generalizability beyond competitive programming. Our setting is competitive programming. Extension to other elite skill markets — mathematics, theoretical computer science, software engineering, professional research — requires either domain-specific natural experiments or cross-domain meta-evidence.

COVID confound. The empirical window straddles the 2020–2021 COVID disruption. Section 5.4 reports COVID-aware sensitivity: the within-individual mechanism evidence and the within-ICPC slope are the cleanest specifications for the AI-shock comparison, the latter because it restricts to contest years 2023 and later, which postdate COVID.

Measurement of AI use and filter strength. We do not directly observe which CF users adopt which AI tools. The cohort $\times$ post DiD identifies practice-pattern shifts at the cohort level; the subpopulation comparison computes a within-user proxy from observed submission-pattern changes. Neither directly measures AI use. The construct-validity case for the AI-prompt signature — discriminant validity against pre-shock skill, and within-user stability around AI release among established users — is developed in Appendix B; Section 5.1 additionally reports that the task-amenability gradient does not discriminate AI from a secular or maturation trend once career stage is held fixed. More fundamentally, the cross-cohort variation in the signature is not identified as AI-caused: the maturation-clean tests in Section 5.1 (the cohort event study, career-time matching, and the difficulty gradient) show a secular practice-norm and novice-maturation component, so the AI reading of the cohort shift is the leading interpretation rather than a demonstrated effect.

The selection-into-AI-prohibited-contests framing rests on the empirical claim that qualification rounds effectively screen out substitute-users; we do not observe filter strength directly. Imperfect proctoring or indirect AI benefit through training would weaken the framework. The within-ICPC

positive slope partially probes filter strength: the positive sign within the qualifier sample is consistent with effective filtering (the residual variation in AI-intensity after the filter selects on complement-users, whose AI-intensity is associated with higher performance); a strong negative sign would indicate the filter is leaky enough that substitute-users contribute meaningfully to within-sample variation. A platform-level or survey-based user-level AI-use measurement would close both identification gaps and is a priority future-work item.

Future research directions. Several extensions follow naturally from the design. First, distinguishing the selection channel from the anticipation channel for AI-prohibited evaluation gates would require a natural experiment in gate introduction or removal, or a longitudinal design tracking training-behavior changes after credible AI-prohibition signals. Second, extension to other elite skill markets where AI-prohibited gates exist (e.g., medical board exams, legal bar exams, scientific peer review) would test the generalizability of the selection mechanism. Third, user-level AI-use measurement through platform telemetry or developer surveys would close the identification gap on individual AI use, sharpening the test of substitution at the individual level. Fourth, a future update incorporating ICPC 2026–2027 and IOI 2026–2027 outcomes would extend the post-shock observation window for the AI-era cohort whose effects are most informative for long-run skill formation.

6.6 Conclusion

We ask whether AI-prohibited evaluation gates can separate two coexisting types of AI practice in elite competitive programming, and we assemble evidence in three patterns. The practice-pattern shift is a cross-cohort move toward the AI-prompt signature on the CF practice panel. The cross-subpopulation asymmetry appears in CF rating gains: among non-rostered CF users (no ICPC/IOI selection), a stronger AI-prompt signature during practice predicts lower contest performance; among roster-linked users (selected through AI-prohibited gates), the relationship becomes null. The within-ICPC result shows that a shift toward AI-style practice predicts *higher* non-AI-aided performance. The same practice input thus carries opposite signs on contest performance depending on whether the contest environment screens. The unified reading is that AI-prohibited qualification gates separate substitute-style from complement-style practice, filtering substitute-users out before they reach elite contests; the selection-into-AI-prohibited-contests structure is the load-bearing feature of the design, and it makes the cross-environment evidence coherent rather than contradictory. The motivating question of our research is whether short-run AI productivity gains carry a long-run cost to elite human-capital formation; our evidence speaks to the gatekeeping institution that would determine whether the long-run cost reaches the elite skill frontier, not to the existence and size of the cost itself. How AI is used within training is one policy lever; the design of the AI-prohibited

evaluation gates that screen access to the elite skill frontier is another. Both levers are worth weighing, as they carry substantial, long-term socioeconomic implications.

A Robustness

This appendix reports the robustness checks referenced in Section 5.4: the ICPC/IOI–CF record linkage and the sensitivity of the cross-subpopulation asymmetry to the excluded name-only matches, the Copilot-anchored difference-in-differences, and the selection-bias robustness for the cross-subpopulation and within-ICPC tests.

A.1 ICPC/IOI–CF linkage and sensitivity to the excluded name-only matches

Candidate competitors from the ICPC and IOI rosters (2018–2025) are linked to CF accounts in two ways. ICPC contestants carry a CF username published on their Competitive Programming Hall of Fame profile, a direct link; the analysis ICPC roster is the 2,189 distinct accounts (2,784 contestant-year records, 88.0% of ICPC contestant-years) that carry such a username. IOI contestants are linked by a name-and-country match to the IOI participant lists, yielding 1,164 accounts. The two sets overlap in 236 accounts, for 3,117 roster-linked CF users.

The name-and-country matcher screens 4,219 candidate persons and returns 1,592 matched to at least one CF username (1,036 ICPC, 467 IOI, 89 both). Each match carries a best-status confidence tier (verified ≥ 0.85 , likely 0.70–0.85, uncertain 0.55–0.70, rejected below 0.55), with counts 441 / 327 / 267 / 557 by person (441 / 338 / 330 / 1,106 across 2,215 match pairs). The matched set spans the competitive frontier: 15 World Champions and 86 gold, 114 silver, and 154 bronze medalists.

The 1,374 users matched to a roster only by a lower-confidence name-and-country match, for which we cannot confirm a contest record or attach a contest score, are excluded from the analysis panel (Section 4.2). The cross-subpopulation asymmetry of Section 5.2 is insensitive to their handling: excluding them (the headline) gives a non-rostered slope of -12.24 and a roster-linked null of -2.41 (pooled interaction -9.82 , $p = 0.048$); leaving them in the non-rostered subpopulation gives -12.16 and -2.41 (interaction -9.75 , $p = 0.048$); and assigning them to the roster-linked subpopulation gives -12.24 and -4.88 (interaction -7.36 , $p = 0.110$). The within-subpopulation slopes are stable across all three handlings; only the pooled-interaction p -value softens when the AI-era-heavy name-only matches are forced into the roster-linked arm, which is itself a reason to prefer their exclusion. Full results are available from the author.

A.2 Robustness: Copilot-anchored DiD

The Section 5.1 headline pins post_t to ChatGPT. A natural question is whether the practice-pattern shift is detectable around the first AI shock (Copilot, 2021-06-29) before ChatGPT was released. We re-run Equation 1 with post_t pinned to the Copilot date and drop the AI-era cohort, whose first rated contest is post-Copilot by construction so the $\text{cohort} \times \text{post-Copilot}$ interaction is unidentified for

them. The Copilot-isolation specification restricts the post window to 2021-Q3 through 2022-Q3, five quarters during which Copilot is the only AI shock ($N = 76,035$ observations across 8,586 users). The transition $\times$ post-Copilot coefficient on `frac_single_shot` is 0.0398 (SE 0.0053, $p < 0.001$), on `median_attempts` is -0.1768 (SE 0.0306, $p < 0.001$), and on `frac_quick_retry` is -0.0100 (SE 0.0033, $p = 0.002$), all three significant in the AI-prompt direction. The transition cohort’s signature shift is therefore already present around the first AI shock, before ChatGPT, so the headline DiD is not a ChatGPT-specific artifact.

The Copilot-isolation quick-retry coefficient (-0.0100 , $p = 0.002$) is also notable on its own: it is negative here (the AI-prompt direction), in contrast to the wrong-signed positive coefficient (0.0090, $p = 0.002$) for the transition cohort in the ChatGPT-anchored headline. This is consistent with the conjecture in Section 5.1: Copilot’s integrated development environment (IDE) autocomplete-style integration may have reduced human retries cleanly, while ChatGPT’s generate-and-debug workflow reintroduced human-debugging cycles among transition users. If anything the shift is more concentrated around Copilot than around ChatGPT for this mid-career cohort: the ChatGPT-anchored transition coefficient on `frac_single_shot` (0.0139 in Table 6) is smaller because its pre-period baseline already absorbs the post-Copilot shift. A second specification that extends the post window through 2025-Q4 ($N = 127,124$) recovers similar magnitudes on single-shot and median-attempts (0.0339 and -0.1569), with the quick-retry coefficient attenuating to null (-0.0027 , SE 0.0028, $p = 0.333$) as the ChatGPT-era observations rejoin the post window.

A.3 Selection-bias robustness for the cross-subpopulation asymmetry test

The cross-subpopulation asymmetry of Section 5.2 (Equations 2 and 3) is estimated on cohort-assigned users active in the post-ChatGPT window: users with at least two post-ChatGPT quarters in 2023-Q1 through 2025-Q4, each containing at least five practice-mode submissions. This filter is required so that `ai_pc1_post` is reliably measured at the user level—a single quarter with one or two submissions produces a wildly noisy first principal component. The cost is sample restriction: 3,729 of the 10,419 cohort-assigned users who pass the Section 5.1 practice-window filter do not pass this activity requirement, dropping the Section 5.2 sample to 6,690 (5,283 non-rostered + 1,407 roster-linked). This appendix documents three robustness checks against selection bias from this restriction: a filter-sensitivity check on the asymmetry test, a cross-section consistency check with the Section 5.1 practice-pattern shift, and a balance check that decomposes the dropped users by observables and by subpopulation.

Filter sensitivity

We re-estimate Equation 2 (per-subpopulation) and Equation 3 (pooled fully-stratified) under three activity thresholds: the headline (≥ 2 quarters, ≥ 5 submissions each), an intermediate variant (≥ 2 quarters, ≥ 1 submission each), and a permissive variant (≥ 1 quarter, ≥ 1 submission). The first principal component is fit once on the headline sample; the resulting PCA loadings and standardization parameters are frozen and applied to the relaxed samples so that `ai_pc1_post` is on the same scale across all three specifications.

Table 11: Cross-subpopulation asymmetry (Equations 2 and 3) under three activity thresholds. PCA loadings frozen on the HEADLINE sample and applied to the relaxed samples. Heteroskedasticity-consistent (HC1) standard errors. The HEADLINE row reproduces Table 7’s θ_1 within numerical noise; small differences arise because the PCA is refit on the Section 5.2 reconstruction sample rather than reusing the broad-panel `ai_pc1_post` scores directly.

Filter	N	θ_{NR}	θ_{RL}	θ_1
PERMISSIVE (≥ 1 q, ≥ 1 sub)	8,478	-14.61*** (1.87)	-7.47* (3.64)	-7.14 ($p = 0.081$)
INTERMEDIATE (≥ 2 q, ≥ 1 sub)	7,753	-16.38*** (2.32)	-4.46 (4.45)	-11.92* ($p = 0.017$)
HEADLINE (≥ 2 q, ≥ 5 subs)	6,690	-12.42*** (2.45)	-2.54 (4.37)	-9.88* ($p = 0.049$)

θ_{NR} is the non-rostered slope on `ai_pc1_post`, θ_{RL} is the roster-linked slope, θ_1 is the pooled cross-subpopulation interaction. Standard errors below each estimate for θ_{NR} and θ_{RL} ; pooled-interaction p -value below θ_1 . * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The pooled interaction θ_1 is significant at HEADLINE and INTERMEDIATE (with INTERMEDIATE actually larger than HEADLINE), and attenuates to marginal significance at PERMISSIVE. The PERMISSIVE attenuation is consistent with classical errors-in-variables: at ≥ 1 quarter with ≥ 1 submission, the per-user signature is computed from very few observations and the resulting measurement noise pulls θ_1 toward zero. Within the non-rostered subpopulation, θ_{NR} is significantly negative across all three filters and ranges from -12.42 to -16.38 . Within the roster-linked subpopulation, θ_{RL} is null at the two stricter thresholds but reaches significance at PERMISSIVE (-7.47 , $p = 0.040$), a pattern we revisit alongside the balance check below.

Cross-section consistency with Section 5.1

If the Section 5.2 sub-sample is materially different from the full Section 5.1 sample in ways that affect the practice-pattern shift, the two analyses are not characterizing the same population, and reading them together is harder to justify. We re-estimate Equation 1 on the 6,690 Section 5.2 users’ user-quarter practice observations in 2018-Q1 through 2025-Q4 (excluding 2022-Q4) and compare to the Section 5.1 headline.

Table 12: Equation 1 AI_era $\times$ post coefficient β_2 on the full Section 5.1 sample (10,419 users) versus the Section 5.2 sub-sample (6,690 users). Cluster-robust (user-level CRV1) standard errors in parentheses.

Outcome	Section 5.1 ($N = 10,419$)	Section 5.2 sub-sample ($N = 6,690$)	Attenuation
frac_single_shot	0.0586*** (0.0063)	0.0509*** (0.0067)	13%
median_attempts	-0.3271*** (0.0374)	-0.2981*** (0.0400)	9%
frac_quick_retry	-0.0190*** (0.0041)	-0.0175*** (0.0043)	8%

*** $p < 0.001$ in all six estimates.

All three primary practice-pattern outcomes survive on the Section 5.2 sub-sample with modest attenuation (between 8% and 13%); statistical significance is preserved in all three. The Section 5.1 and Section 5.2 findings are about the same underlying population: the sub-sample restriction does not select on the practice-pattern signal in a way that would explain the cross-subpopulation asymmetry by sample composition alone.

Balance check: in-sample versus dropped users

The 3,729 users dropped at the activity filter are compared to the 6,690 retained users on pre-ChatGPT mean CF rating, Δ rating (post – pre), the spanning indicator (whether the user is active on both sides of the ChatGPT cutoff), the pre-window single-shot AC rate (a proxy for pre-window AI signature among the subset of dropped users with at least one pre-window practice submission), and cohort composition.

Table 13: Continuous observables: Section 5.2 in-sample users versus dropped users. Mean $\pm$ standard error of the mean; Welch t -test p -values.

Observable	In-sample	Dropped	Diff	p
Pre-ChatGPT mean CF rating	1,459 $\pm$ 5 ($n = 6,690$)	1,642 $\pm$ 7 ($n = 3,286$)	-183	< 0.0001
Δ rating (post – pre)	300 $\pm$ 3 ($n = 6,690$)	92 $\pm$ 3 ($n = 3,286$)	208	< 0.0001
Spanning indicator	0.79 $\pm$ 0.01 ($n = 6,690$)	0.30 $\pm$ 0.01 ($n = 3,729$)	0.50	< 0.0001
Pre-window single-shot AC rate	0.45 $\pm$ 0.00 ($n = 5,952$)	0.42 $\pm$ 0.01 ($n = 1,894$)	0.03	< 0.0001

Table 14: Cohort composition: Section 5.2 in-sample users versus dropped users. Chi-square test of independence $p < 0.0001$.

Cohort	In-sample %	Dropped %	In count	Drop count
pre_AI	24.2%	46.4%	1,616	1,731
transition	57.7%	38.4%	3,861	1,432
AI_era	18.1%	15.2%	1,213	566

The dropped users are not low-skill quitters; they are higher-skilled veterans. Pre-ChatGPT mean CF rating averages 1,642 for the dropped group versus 1,459 for the in-sample group (a 183-point gap), with much smaller post-shock rating gains (92 versus 300) consistent with mean reversion at higher rating levels. They are predominantly non-spanning (30% versus 79%), reflecting that many are users who reduced activity after ChatGPT rather than users who exited entirely. The pre-window single-shot AC rate is only marginally different across groups (0.42 dropped versus 0.45 in-sample), suggesting that dropout is not strongly correlated with pre-window AI signature. The cohort composition is sharply different: the pre-AI cohort is over-represented among dropped users (46.4% versus 24.2%), consistent with older veterans aging out of CF practice after ChatGPT.

Table 15: Differential dropout at the activity filter, by subpopulation.

Subpopulation	N at Section 5.1 sample	N at Section 5.2 sample	Dropout %
roster-linked	2,480	1,407	43.3%
non-rostered	7,939	5,283	33.5%

The most consequential finding from the balance check is the asymmetric dropout between the two subpopulations. Roster-linked users (those with ICPC or IOI affiliation) drop out at the activity filter more often (43.3%) than non-rostered users (33.5%). This is the opposite direction from the naive hypothesis that roster-linked users are more committed and therefore more consistently active. The likely substantive cause is that ICPC and IOI contestants compete in time-bound events and tend to step back from open Codeforces practice after their qualifying year ends; non-rostered users, who chose CF as their primary competitive outlet, are more consistently active across years.

The implication for the asymmetry test: the roster-linked sample is the more-selected of the two subpopulations, and the null roster-linked slope ($\theta_{RL} = -2.41$ in Table 7) could be partially produced by differential censoring of high-signature roster-linked users rather than purely by the AI-prohibited-qualifier screening mechanism. The filter-sensitivity result in Table 11 above is consistent with this reading: when the PERMISSIVE filter brings additional low-density roster-linked users into the sample, θ_{RL} becomes significantly negative (-7.47 , $p = 0.040$). The asymmetry test θ_1 should be interpreted as bounded by this differential censoring; the cross-environment opposite-signs pattern with the within-ICPC test (Section 5.3) remains the more robust evidence of the screening-filter mechanism.

Skill-control sensitivity

The headline specification includes the baseline-skill control $\bar{r}_h^{\text{pre}}$ to block omitted-variable bias from the negative AI-signature-skill correlation (Section 5.2). Dropping the control shifts the within-subpopulation slopes substantially: non-rostered moves from -12.24 to -5.15 ($p = 0.050$),

roster-linked from -2.41 to 4.88 ($p = 0.29$). The pooled slope-difference θ_1 is essentially preserved at -10.04 (SE 5.29 , $p = 0.058$). Because the AI-signature-skill correlation and the mean-reversion rate are similar in both subpopulations, the omitted-variable bias shifts within-subpopulation levels by similar amounts and largely cancels in the cross-subpopulation difference. The cross-subpopulation asymmetry test does not depend on the skill-control specification, even though the within-subpopulation levels do.

Roster-skill confound check

The cross-subpopulation asymmetry could in principle reflect a roster-versus-skill confound: roster-linked users are systematically higher-skill, having qualified for ICPC and IOI rosters through competitive selection, so a story in which higher-skill users have systematically different post-shock rating dynamics could in principle produce the asymmetry without any AI-specific channel. To rule out the simplest version of this confound, we recompute the three AI-signature components on contest submissions instead of practice submissions and re-run the within-mode roster comparison. The bootstrap-validated contest-mode PCA loadings are within 0.05 of the practice-mode loadings on every component ($0.653, 0.551, 0.520$ against the practice-mode $0.602, 0.570, 0.560$), so the composite construct is stable across modes.

If the practice-mode roster gap were entirely a roster-skill artifact, the contest-mode gap should attenuate with the skill control in the same direction the practice-mode gap does. The empirical pattern is the opposite. Without $\bar{r}_h^{\text{pre}}$, the practice-mode roster gap is -0.256 ($p < 0.001$, $N = 7,585$) and the contest-mode roster gap is -0.226 ($p < 0.001$, $N = 7,639$). Adding $\bar{r}_h^{\text{pre}}$ attenuates the practice-mode gap to -0.168 ($p < 0.001$, $N = 7,151$), as a roster-skill story would predict, but it *widens* the contest-mode gap to -0.262 ($p < 0.001$, $N = 7,191$). The roster-skill story predicts both gaps shrink together; the data show one shrinks and the other grows. The roster gap on the contest-mode signature, conditional on baseline skill, is therefore robust to the simplest version of the confound.

The contest-mode and practice-mode signatures are correlated with $r \approx 0.32$ at the user level (identical across roster-linked and non-rostered subpopulations to three decimal places). The two measures share roughly ten percent of variance: enough that the contest-mode test is not orthogonal to the practice-mode test, but not so much that contest mode is a tautological re-test. The contest-mode evidence is therefore partial corroboration of the skill-confound rule-out, not direct measurement of in-contest AI use. A non-rostered user can show high contest signature by actually using AI during CF contests (which is hard to police), by carrying learned reflexes from AI-heavy practice into contest mode, or by producing faster but lower-quality submissions because deliberative debugging skill has atrophied. The contest-mode test cannot separate these channels.

A.4 Robustness for the within-ICPC AI-intensity test

The within-ICPC AI-intensity test of Section 5.3 (Equation 4) is estimated on the 444-user, 579-observation sample described in Table 9. The headline $\psi_{AI} = 0.190$ ($p = 0.028$) depends on the treatment construct (within-user changes vs post-window levels) and on the activity filter that defines the qualifying practice window. This appendix documents four robustness exhibits: filter sensitivity, construct sensitivity, a balance check on filter dropout by cohort, and a Section 5.1 cross-check on the within-ICPC sub-sample.

Filter sensitivity

We re-estimate Equation 4 under three activity-filter thresholds for measuring within-user changes in the three behavioral proxies. The treatment is built per (user, contest year), matching the contest-year-conditional window logic of the main spec but with a faithfully per-(user, year) aggregation (the main spec’s per-user first-year simplification, which collapses each user to one treatment value, is treated separately in the construct exhibit below). PCA loadings are fit on the headline-threshold sample and frozen across the moderate and least-strict thresholds so that the treatment is on a comparable scale across thresholds.

Table 16: Within-ICPC AI-intensity test under three activity-filter thresholds, with the treatment built per (user, contest year) and aggregated as within-user changes. $AI_{era} \times ai_intensity$ coefficient with team-clustered standard errors. PCA loadings are fit on the headline-threshold sample and frozen.

Filter	N_{obs}	N_{users}	ψ_{AI}	SE	p
LOOSE (≥ 1 quarter, ≥ 1 sub)	541	465	0.434	0.257	0.091
MEDIUM (≥ 2 quarters, ≥ 1 sub)	405	355	0.162	0.246	0.512
HEADLINE (≥ 2 quarters, ≥ 5 subs each)	386	336	0.108	0.213	0.611

Direction is positive across all three thresholds; magnitude varies from 0.108 to 0.434; none reaches conventional significance (smallest $p = 0.091$ at the least-strict threshold). The headline-threshold row at 0.108 does not match the manuscript’s 0.190 because the manuscript spec uses per-user first-year changes (collapsing each user’s per-(user, year) treatment to a single per-user value) and fits the PCA on the broader CF qualifying reference group rather than the within-ICPC sub-sample; the construct exhibit below isolates that dependence.

Construct sensitivity

The manuscript treatment is PCA on within-user changes (first differences) with per-user first-year aggregation. Three alternative constructs change either the levels-vs-changes choice or the per-user-

vs-per-(user, year) aggregation. All four constructs use the same headline activity filter; PCA is refit per construct.

Table 17: Within-ICPC construct sensitivity: $AI_era \times ai_intensity$ coefficient under alternative treatment definitions.

Construct	N_{obs}	ψ_{AI}	p
Per-user first-year changes (manuscript)	579	0.190	0.028
Per-(user, year) changes	386	0.108	0.611
Per-user levels (<code>ai_pc1_post</code> analog, full post window)	995	0.059	0.422
Per-(user, year) levels	525	-0.023	0.781

Only the manuscript construct (per-user first-year changes) reaches significance. Direction is positive in three of four constructs; magnitudes span -0.023 to 0.190 . The within-user first-difference construct is theoretically motivated for a skill-formation hypothesis because it nets out each user’s pre-shock baseline practice style, but readers should treat the 0.190 magnitude as construct-dependent rather than as a tight point estimate.

Balance check on filter dropout by cohort

Step 3 of Table 9 drops 579 of the 1,026 ICPC participants at Step 2, retaining 447. The first-difference construct is stricter than a post-window-levels filter because it requires both a pre-window (2021-Q3 to 2022-Q3) and a post-window (2023+) with the qualifying activity thresholds. Dropout rates differ substantially across cohorts:

Table 18: Within-ICPC sample dropout at Step 3 of Table 9 (`ai_intensity_pc1` defined under the within-user first-difference construct), by cohort.

Cohort	N at Step 2	N at Step 3	Dropped	Dropout %
pre_AI	431	144	287	66.6%
transition	412	233	179	43.4%
AI_era	183	70	113	61.7%

Transition users drop out the least (43.4%), pre_AI users the most (66.6%); AI_era (61.7%) sits between. The pattern is consistent with the first-difference construct’s both-windows requirement: pre_AI users are more likely to have stopped active CF practice by 2023, AI_era users are more likely to lack pre-2022 practice history (their rated career started post-Copilot by construction), and transition users span both periods most fully. Dropped users have higher pre-ChatGPT mean rating (1,932 vs 1,802, Welch $p < 0.0001$) and lower spanning rate (0.74 vs 0.85, $p < 0.0001$). ICPC score itself does not differ between retained and dropped users ($p = 0.73$), so the dropout is not directly outcome-correlated; pre-window single-shot AC rate is modestly higher among retained

users (0.46 vs 0.42, $p < 0.001$). The cohort interaction in the headline is identified off a sample whose AI_era cell is the second-most-attributed; whether this biases the AI_era $\times$ ai_intensity slope upward or downward depends on the within-cohort relationship between the within-user change in AI signature and dropout, which we do not directly observe.

Cross-check with the practice-pattern shift

If the within-ICPC sub-sample participates in the broader practice-pattern shift documented in Section 5.1, the AI_era $\times$ post DiD coefficients on the three behavioral proxies should survive on these users' user-quarter observations. We re-run Equation 1 on the 446 users that the reconstruction retains at Step 4 (within ± 2 of the headline 444-user sample, an acceptable reconstruction tolerance).

Table 19: Equation 1 AI_era $\times$ post coefficient β_2 on the full Section 5.1 sample (10,419 users) versus the within-ICPC sub-sample (446 users). Cluster-robust (user-level) standard errors in parentheses.

Outcome	Full sample (10,419 users)	Within-ICPC sub-sample (446 users)
frac_single_shot	0.0586 (0.0063, $p < 0.001$)	0.0736 (0.0195, $p < 0.001$)
median_attempts	-0.3271 (0.0374, $p < 0.001$)	-0.2705 (0.1130, $p = 0.017$)
frac_quick_retry	-0.0190 (0.0041, $p < 0.001$)	-0.0193 (0.0134, $p = 0.15$)

The single-shot effect is *amplified* on the within-ICPC sub-sample to 126% of its full-panel magnitude (0.0736 vs 0.0586, $p < 0.001$). The median-attempts effect retains 83% of full (-0.2705 vs -0.3271, $p = 0.017$). The quick-retry effect has essentially the same magnitude as the full panel (-0.0193 vs -0.0190) but loses statistical significance on the smaller sample ($p = 0.15$). The amplified pattern is consistent with the within-ICPC sub-sample being more-active by construction (the first-difference construct requires qualifying practice in both the 2021-Q3 to 2022-Q3 pre-window and the 2023+ post-window), so the practice-pattern shift is more strongly identified than on the full panel.

B Construct Validation of the AI-Prompt Signature

The headline mechanism results (Section 5.1) and the individual-level qualifier-stage test rest on the AI-prompt signature: the behavioral composite of higher single-shot accept rate, lower median attempts-to-AC, and lower quick-retry rate that we attribute to AI prompt-and-fix substitution. A skeptical reading is that this signature captures programmer skill or an analytic work style rather than AI use directly. This appendix bounds that concern with three tests on the Codeforces panel restricted to submissions where `author.participantType == "PRACTICE"` (the canonical practice filter on the combined re-scraped panel; each test below reports its own sample).

The tests address two distinct alternatives to the AI-substitution interpretation. Subsection B.1 tests whether the signature is a proxy for skill, by examining its relationship to CF rating in the strictly pre-shock window (2018-Q1 to 2022-Q3). Subsection B.2 examines whether established practitioners exhibit a within-user discontinuity in the signature at AI release dates, which would distinguish a discrete AI-availability effect from a smooth skill-acquisition trend. Subsection B.3 synthesizes the two tests against the experimental causal anchor from Shen and Tamkin (2026).

B.1 Discriminant validity against pre-shock skill

If the signature were a tight proxy for CF rating, pre-2022 high-rated users would already exhibit the AI-prompt signature: higher single-shot accept rates, lower median attempts to AC, and lower quick-retry rates than lower-rated users. We test this directly using the pre-shock window 2018-Q1 through 2022-Q3 (strictly before ChatGPT, which released 2022-11-30).

Cross-sectional test. For each user with at least four active pre-shock quarters ($n = 8,535$), we compute the user-level mean of each signature component and the user-level mean pre-shock CF rating. Pearson and Spearman correlations between the signature components and mean rating are reported in Table 20.

Table 20: Pre-shock cross-sectional correlation: signature components vs. CF rating

Signature component	n users	Pearson r	SE	p	Spearman r
Single-shot AC rate	8,535	-0.091	0.011	< 0.001	-0.095
Median attempts to AC	8,535	+0.157	0.011	< 0.001	+0.175
Quick-retry rate	8,535	+0.147	0.011	< 0.001	+0.161

Notes. Pre-shock window: 2018-Q1 to 2022-Q3. User-level means; users included if active in ≥ 4 pre-shock quarters with ≥ 5 practice submissions per quarter. Filter: `author.participantType == "PRACTICE"` on the combined re-scraped panel.

All three correlations have signs *opposite* to the AI-prompt prediction. In the pre-shock window, higher-rated users had *lower* single-shot AC rates, *higher* median attempts to AC, and higher quick-retry rates than lower-rated users. Skill in the pre-shock era was associated with attempting

harder problems that required more iteration, not with the prompt-and-fix pattern. Magnitudes are weak ($|r| < 0.20$) but the directional contrast with the AI-prompt prediction is unambiguous.

Figure 3 plots the user-level scatter and binned mean for each component against pre-shock rating. The binned-mean lines make the directional contrast visible.

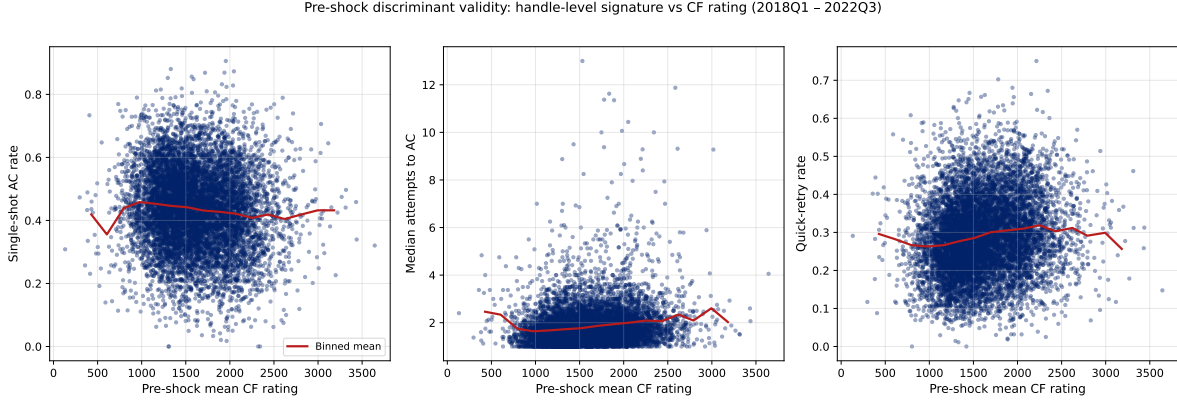


Figure 3: Pre-shock discriminant validity: user-level mean signature components vs. pre-shock mean CF rating ($n = 8,535$ users, 2018-Q1 to 2022-Q3). Red lines are binned means. The cross-sectional relationships have signs opposite to the AI-prompt prediction: higher pre-shock rating is associated with lower single-shot AC, higher median attempts, and higher quick-retry rates — inconsistent with a skill-proxy interpretation of the signature.

Within-user dynamics. The cross-sectional contrast could in principle mask a within-user skill-on-signature effect. We test this with the specification

$$\text{signature}_{ht} = \alpha_h + \gamma_t + \delta \cdot \frac{\text{rating}_{ht}}{100} + \varepsilon_{ht}, \quad (5)$$

where α_h is a user fixed effect, γ_t is a quarter fixed effect, and δ is the within-user elasticity per 100 Elo points. The sample is the pre-shock quarterly panel; standard errors are clustered at the user level. Table 21 reports the results.

Table 21: Pre-shock within-user elasticity: signature vs. rating				
Signature component	n obs	δ per 100 Elo	SE	p
Single-shot AC rate	80,184	+0.00434	0.00045	< 0.001
Median attempts to AC	79,514	−0.03335	0.00367	< 0.001
Quick-retry rate	80,203	+0.00110	0.00036	0.002

Notes. Spec (5): user and quarter FE, cluster-robust SE at user. Pre-shock window: 2018-Q1 to 2022-Q3.

Within-user, a user’s signature does shift modestly with its own rating, but the magnitudes are small. The single-shot AC elasticity is $\delta = 0.0043$ per 100 Elo, meaning that a user would need to gain roughly 1,350 Elo points to produce a single-shot AC shift of magnitude comparable to the

Pattern A AI-era cohort $\times$ post effect of +0.0586 documented in Section 5.1. Median attempts has an elasticity of -0.033 per 100 Elo against the Pattern A effect of -0.327 , roughly a 10:1 ratio of cohort-effect to skill-elasticity. Skill movements at the magnitudes plausibly achievable within the 2022-Q4 to 2025 post-shock window (typically tens to low hundreds of Elo for established users, not 1,000+) cannot reproduce the cohort effect.

Taken together, the cross-sectional and within-user tests establish that the signature is not a tight proxy for skill: the cross-sectional sign runs opposite to the AI-prompt prediction, and the within-user skill-on-signature elasticity is an order of magnitude too small to explain the Pattern A cohort effect.

B.2 Within-user stability around AI release

The remaining alternative for the Pattern A cohort effect is a smooth-trend story: the signature has been drifting toward the AI-prompt signature over time for reasons unrelated to AI release, and we mistakenly attribute the cohort contrast to AI exposure. A direct test is whether established users exhibit a discrete change in their signature at known AI release dates.

We estimate a within-user event study around the ChatGPT release (2022-11-30) on the spanning panel of users active both pre- and post-shock ($n = 1,712$ users, 25,395 user-month observations):

$$\text{signature}_{ht} = \alpha_h + \sum_{k=-18, k \neq -1}^{+18} \beta_k \cdot \mathbf{1}[\text{event}_{ht} = k] + \varepsilon_{ht}, \quad (6)$$

where event time is months from 2022-11. User FE absorbs user-level heterogeneity; the reference period is one month before release. Standard errors are clustered at the user level. Figure 4 plots the coefficient time series.

The estimated coefficient profile shows no discrete shift at $k = 0$. The primary outcome (single-shot AC rate) has coefficients clustered near zero across the pre- and post-shock window, with a few scattered significant points that do not form a coherent post-shock pattern. The same holds for the secondary outcomes (median attempts to AC, quick-retry rate; details available from the author).

The Copilot release (2021-06-29) yields a similar conclusion with additional pre-shock parallel-trends concerns (pre-shock leads are scattered significantly positive on the primary outcome). We treat the Copilot event study as inconclusive due to the parallel-trends violation and report it for completeness; full results are available from the author.

Cohort-stratified event studies on the ChatGPT shock confirm the absence of within-user adoption among existing practitioners. Within the pre-AI cohort ($n = 733$ users, 8,791 user-month obs in the event window), the signature is essentially flat across event time. Within the transition cohort

($n = 775$ users, 14,460 obs), the post-shock pattern is in the same direction as the pre-AI cohort, with no discrete jump at $k = 0$.

Interpretation. Established practitioners did not shift their practice patterns at ChatGPT release. Two implications follow. First, the smooth-trend alternative is bounded: if a smooth time trend were driving the Pattern A cohort contrast, established users would also drift toward the AI-prompt signature over time, and they do not. Second, the Pattern A cohort effect must originate primarily in cohort *composition*: users who entered the platform after AI’s broad availability arrived with practice patterns that differ from pre-AI cohorts’. This is consistent with the paper’s skill-formation framework: AI exposure during the formation period shapes practice habits; AI exposure to experienced practitioners does not change theirs.

B.3 Synthesis with experimental evidence

The two tests bound two distinct alternatives to the AI-substitution interpretation. The discriminant test (Subsection B.1) rules out the signature as a proxy for *static* skill: the cross-sectional sign runs opposite to the AI-prompt prediction and the within-user skill-on-signature elasticity is an order of magnitude too small to explain the Pattern A cohort effect. The event-study test (Subsection B.2) rules out adoption by established practitioners: there is no within-user drift in the signature around the AI releases, which locates the Pattern A cohort effect in cohort composition rather than in existing-user adoption.

What these two tests establish is narrower than AI causation: the cross-cohort signature difference is a genuine, cohort-compositional shift in practice style, not an artifact of static skill or of established users adopting AI. They do not establish that the cohort difference is *caused* by AI. As Section 5.1 shows, the cohort gradient does not coincide with the AI releases (the cohort event study rejects parallel trends and shows no discontinuity), the AI-era and transition cohorts are indistinguishable at matched career stage, and the maturation-clean difficulty gradient is skill-like rather than AI-amenable. The signature is therefore best read as a practice-style proxy whose cross-cohort variation carries a secular and maturation component; the AI interpretation is the leading reading, not an identified effect.

Shen and Tamkin (2026)’s randomized experiment on 52 software developers provides the complementary causal anchor. Their RCT randomly assigns AI use during a learning task and finds that AI users score substantially lower on a subsequent non-AI-aided assessment (a 17% reduction, Cohen’s $d = 0.738$, $p = 0.010$), with no significant gain in completion time; the deficit is concentrated among low-engagement, prompt-and-fix usage patterns rather than the cognitively engaged ones. That low-engagement mode of working is the behavioral counterpart to the AI-prompt signature we measure; because AI use is randomly assigned in their design, its link to weaker

non-AI-aided performance is causally identified in that setting. Our signature is the multi-year, field-observational analog of their experimentally induced pattern.

Taken together, the field tests and the experimental anchor establish that the AI-prompt signature is a behavioral practice-style proxy consistent with AI-assisted practice: the discriminant and event-study tests rule out static skill and existing-user adoption, and [Shen and Tamkin \(2026\)](#)’s RCT shows that AI use produces this very pattern. What they do not establish is that the cross-cohort variation in the signature is AI-caused; the maturation-clean evidence in [Section 5.1](#) attributes part of it to secular practice-norm change and novice maturation. We therefore use the signature as a proxy for AI-style practice throughout, and treat its AI interpretation as the leading but unverified reading rather than a measured fact.

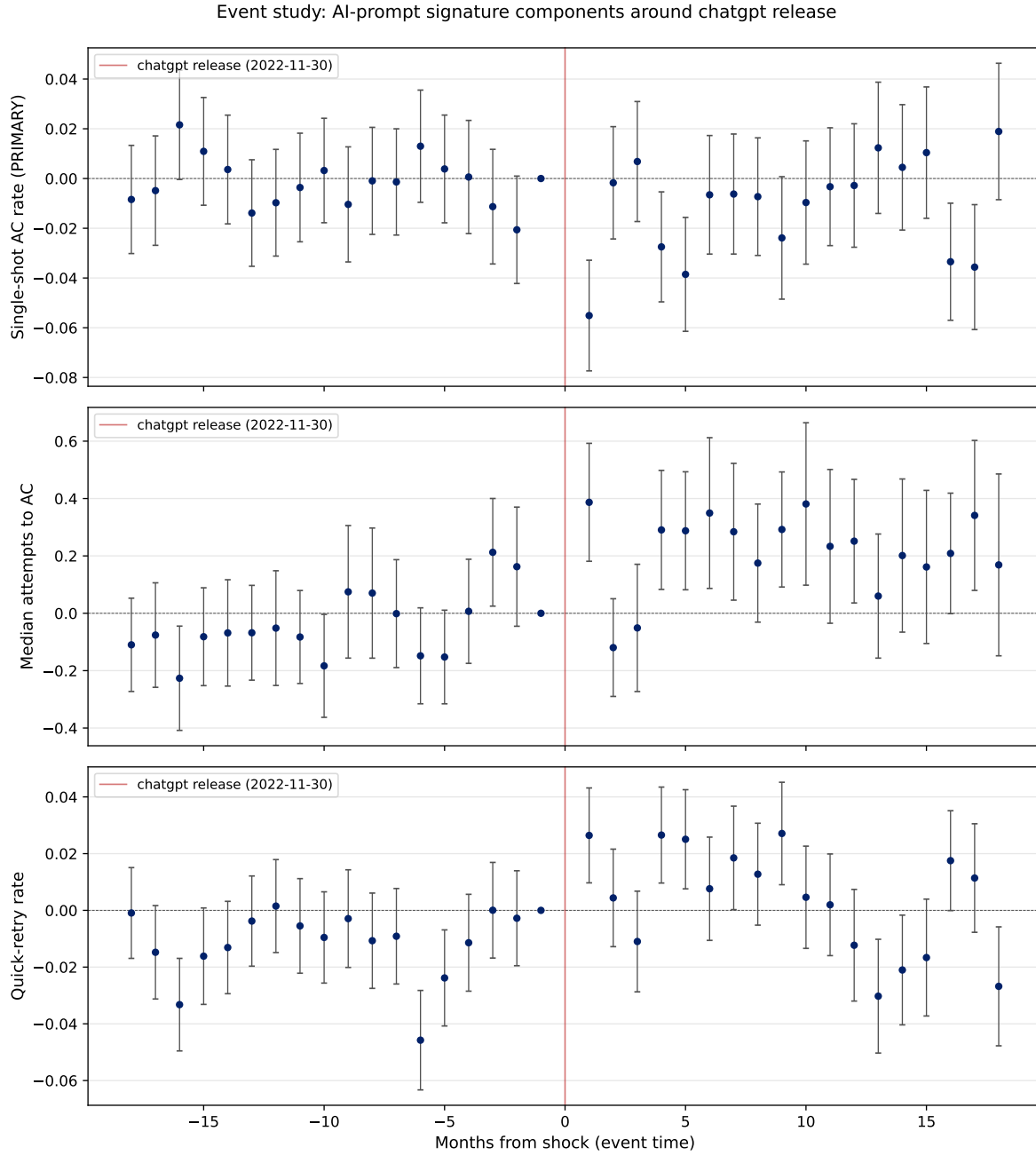


Figure 4: Within-user event study at the ChatGPT release. Spanning panel of 1,712 users active both pre- and post-shock. Reference period: $k = -1$ (one month before release, October 2022). Vertical red line marks release. No discrete jump at $k = 0$ in any of the three signature components. Pre-shock leads are jointly indistinguishable from zero; post-shock coefficients are mostly null or scattered, with no monotone pattern.

C Data and code availability

CF rating and submission data: scraped from the CF API (real usernames; three-stage procedure in Section 4.2). ICPC standings: scraped from cphof.org. IOI results: scraped from stats.ioinformatics.org. The analysis code and cleaned panels, together with the scoring-system specifications (CF Elo, ICPC team scoring, IOI individual scoring), are available from the author on request. A de-identified replication package will accompany the published article.

References

- Acemoglu, D., D. Kong, and A. Ozdaglar (2026, February). AI, Human Cognition and Knowledge Collapse. NBER Working Paper 34910, National Bureau of Economic Research.
- Bastani, H., O. Bastani, A. Sungu, H. Ge, Ö. Kabakcı, and R. Mariman (2025). Generative AI Without Guardrails Can Harm Learning: Evidence from High School Mathematics. *Proceedings of the National Academy of Sciences* 122(26), e2422633122.
- Becker, G. S. (1964). *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. National Bureau of Economic Research, Columbia University Press.
- Brynjolfsson, E., D. Li, and L. Raymond (2025). Generative AI at work. *Quarterly Journal of Economics* 140(2), 889–942.
- Cinelli, C. and C. Hazlett (2020). Making Sense of Sensitivity: Extending Omitted Variable Bias. *Journal of the Royal Statistical Society Series B* 82(1), 39–67.
- Cunha, F., J. J. Heckman, and S. M. Schennach (2010). Estimating the Technology of Cognitive and Noncognitive Skill Formation. *Econometrica* 78(3), 883–931.
- Gaessler, F. and H. Piezunka (2023). Training with AI: Evidence from chess computers. *Strategic Management Journal* 44(11), 2724–2750. Pre-generative-AI counterfactual; chess-computer access raised player Elo by approximately 11 points per year.
- Hausman, N., O. Rigbi, and S. Weisburd (2025, April). Generative AI’s Impact on Student Achievement and Implications for Worker Productivity. CESifo Working Paper 11843, CESifo. Also CEPR Discussion Paper 20206; SSRN 5224465.
- Hoffmann, M., S. Boysel, F. Nagle, S. Peng, and K. Xu (2024). Generative AI and the Nature of Work. Working Paper 25-021, Harvard Business School. SSRN 5007084.
- Lee, D. S. (2009). Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects. *Review of Economic Studies* 76(3), 1071–1102.
- Markus, M. L. and M. S. Silver (2008). A Foundation for the Study of IT Effects: A New Look at DeSanctis and Poole’s Concepts of Structural Features and Spirit. *Journal of the Association for Information Systems* 9(10/11), 609–632. Functional affordances framework in IS.

- Melumad, S. and J. H. Yun (2025). Experimental evidence of the effects of large language models versus web search on depth of learning. *PNAS Nexus* 4(10), pgaf316.
- Mincer, J. (1974). *Schooling, Experience, and Earnings*. National Bureau of Economic Research, Columbia University Press.
- Mirzayanov, M. (2015, October). Open Codeforces Rating System. Codeforces blog entry 20762. Documents the modified Elo rating system used by Codeforces.
- Orlikowski, W. J. (2000). Using Technology and Constituting Structures: A Practice Lens for Studying Technology in Organizations. *Organization Science* 11(4), 404–428. Technologies-in-practice: same artifact under common access produces systematically different enacted use patterns across user groups.
- Oster, E. (2019). Unobservable Selection and Coefficient Stability: Theory and Evidence. *Journal of Business & Economic Statistics* 37(2), 187–204.
- Shen, J. H. and A. Tamkin (2026). How AI Impacts Skill Formation. *arXiv preprint*.
- Shihab, M. I. H., C. Hundhausen, A. Tariq, S. Haque, Y. Qiao, and B. Mulanda (2025). The Effects of GitHub Copilot on Computing Students’ Programming Effectiveness, Efficiency, and Processes in Brownfield Programming Tasks. In *Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER 2025)*.
- Spence, M. (1973). Job Market Signaling. *Quarterly Journal of Economics* 87(3), 355–374.
- Stadler, M., M. Bannert, and M. Sailer (2024). Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior* 160, 108386.
- Stiglitz, J. E. (1975). The Theory of “Screening,” Education, and the Distribution of Income. *American Economic Review* 65(3), 283–300.